

Network Layer



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Outline

- IPv4 Addressing, CIDR
- Sub-netting and Super-netting
- IPv6
- IP Datagram (IPv4) / IP Protocol
- NAT
- Routing protocol: DV routing and LS routing
- Hierarchical Routing
- Internet Routing: RIP, OSPF and BGP
- Broadcast Routing and Multicast routing
- ARP, RARP, and DHCP

IPv4 Addresses

- An IP address is a **32-bit address**.
- The address space of IPv4 is 2^{32} which is equal to 4,294,967,296 (more than 4 billion).
- The IP addresses are **unique**.
 - Each address defines one, and only one, **connection to the network**.
- Is it possible for a device to have **two IP addresses**?
 - If a device has two connections to the internet, **via two networks**, it has **two IP address**.

IPv4 Addresses

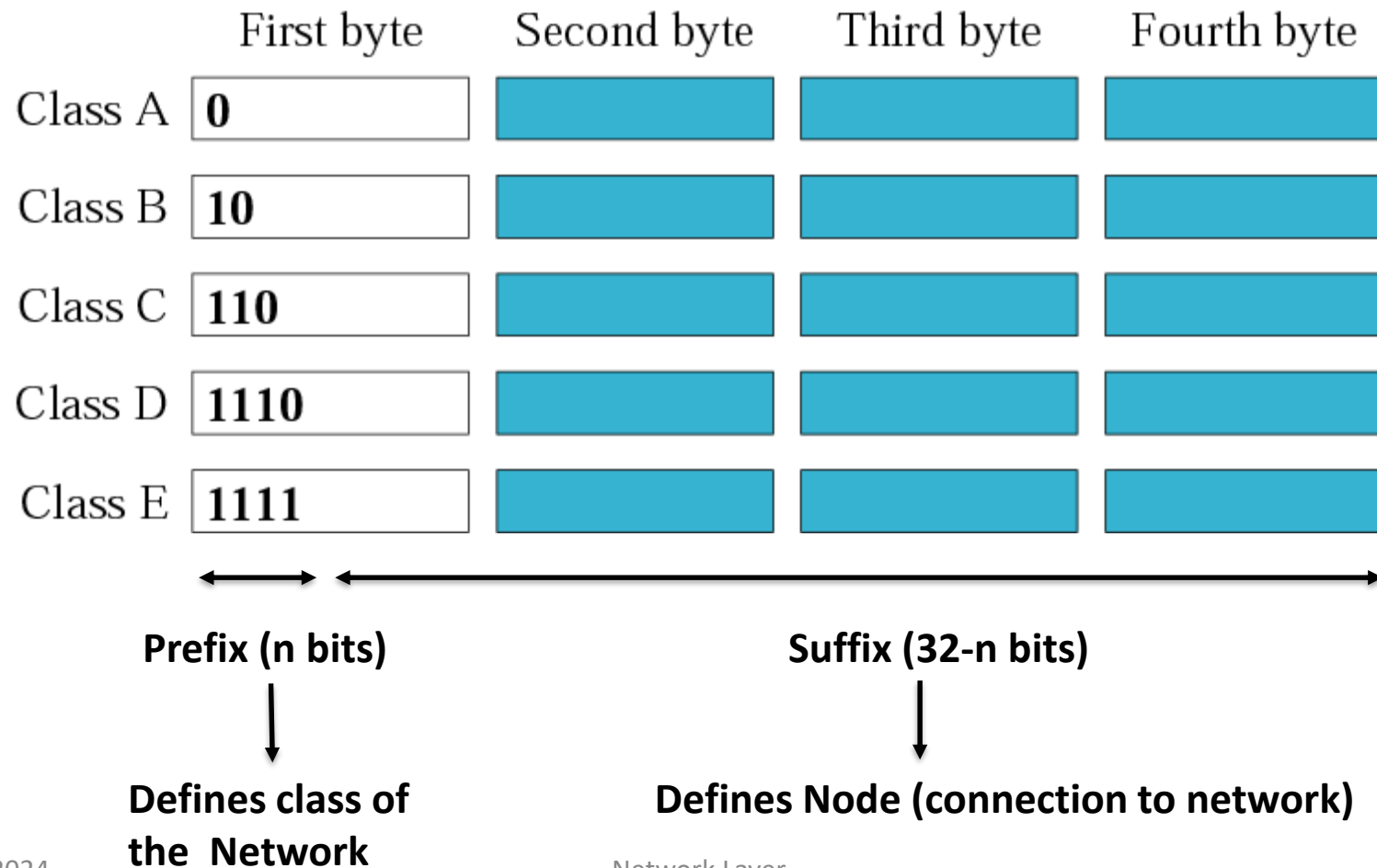
- Classful addressing (**Addresses possible in each class**)
 - Class A : 2^{31} or 2,147,483,648
 - Class B : 2^{30} or 1,073,741,824
 - Class C : 2^{29} or 536,870,912
 - Class D: 2^{28} or 268,435,456
 - Class E : 2^{28} or 268,435,456

Occupation of the address space

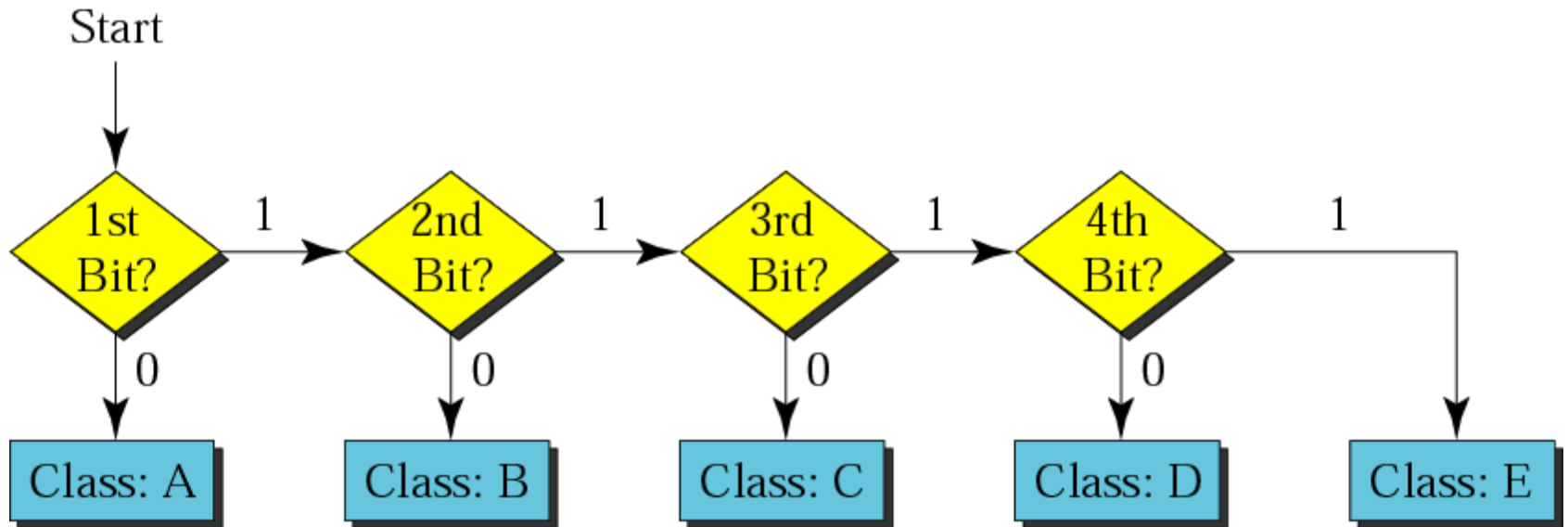
Address space



Finding the class in **binary** notation [5]



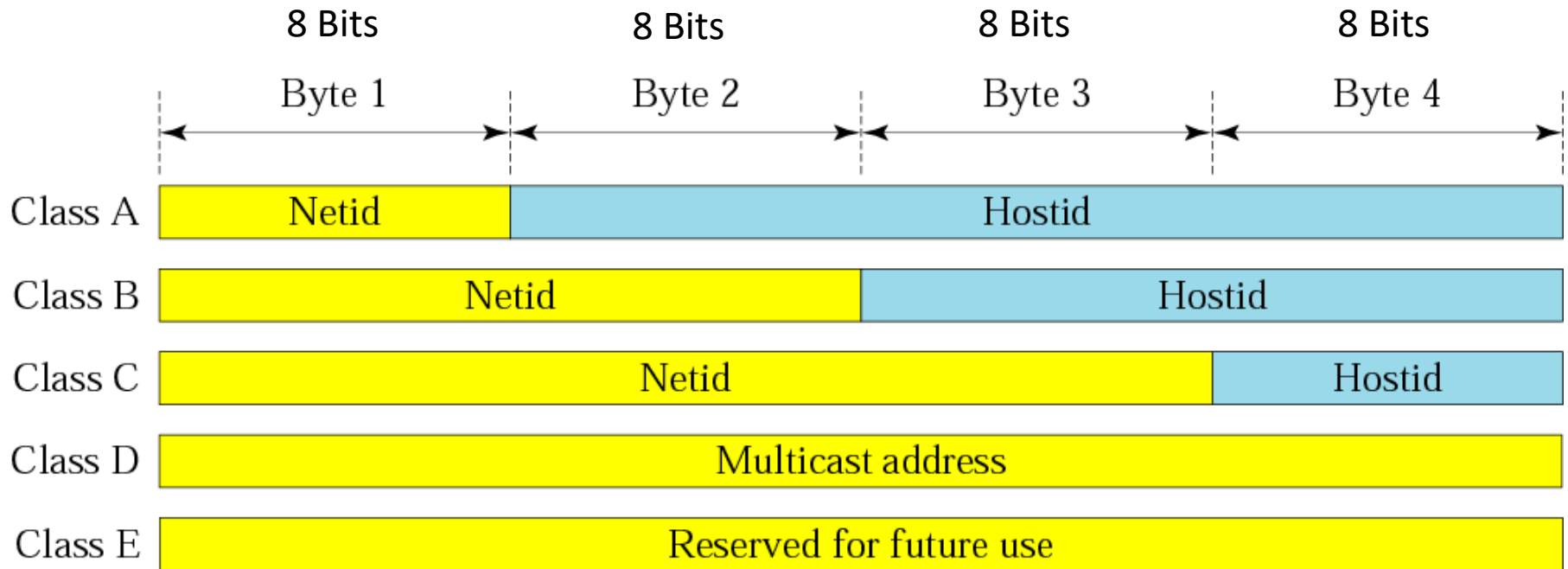
Finding the address class [5]



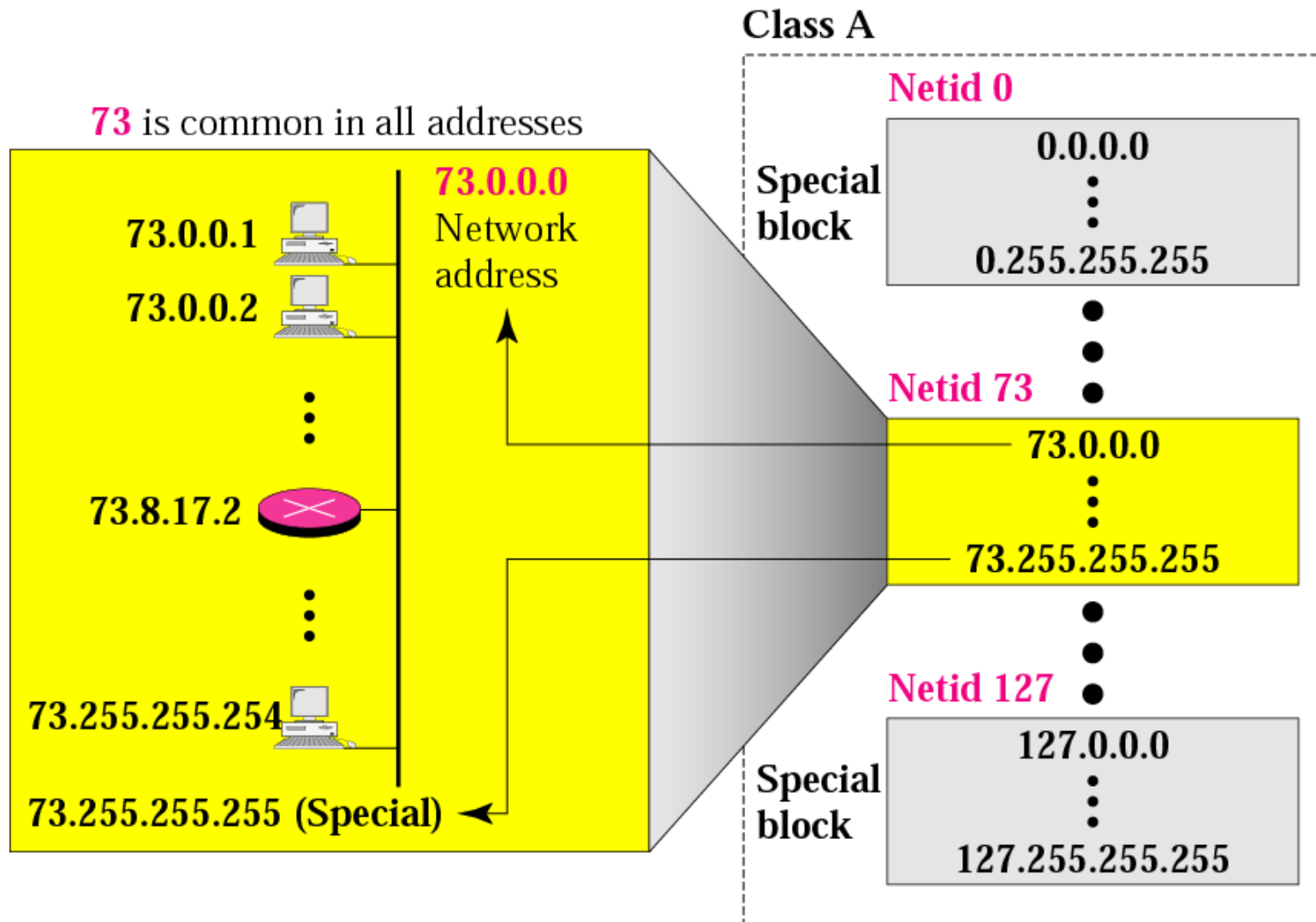
Finding the class in decimal notation[5]

	First byte	Second byte	Third byte	Fourth byte
Class A	0 to 127			
Class B	128 to 191			
Class C	192 to 223			
Class D	224 to 239			
Class E	240 to 255			

Netid and hostid [5]

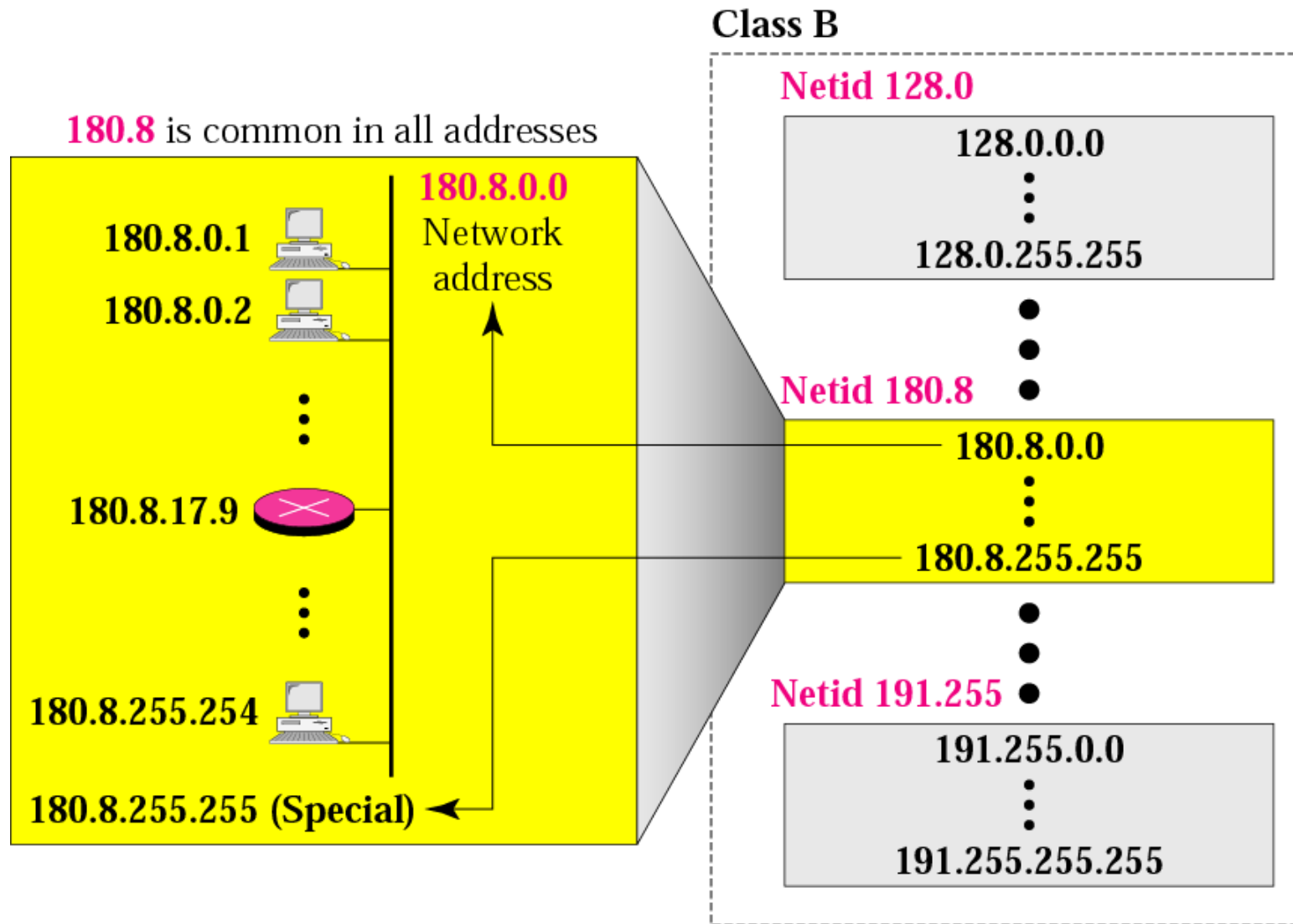


Blocks in class A[5]



128 blocks: 16,777,216 addresses in each block

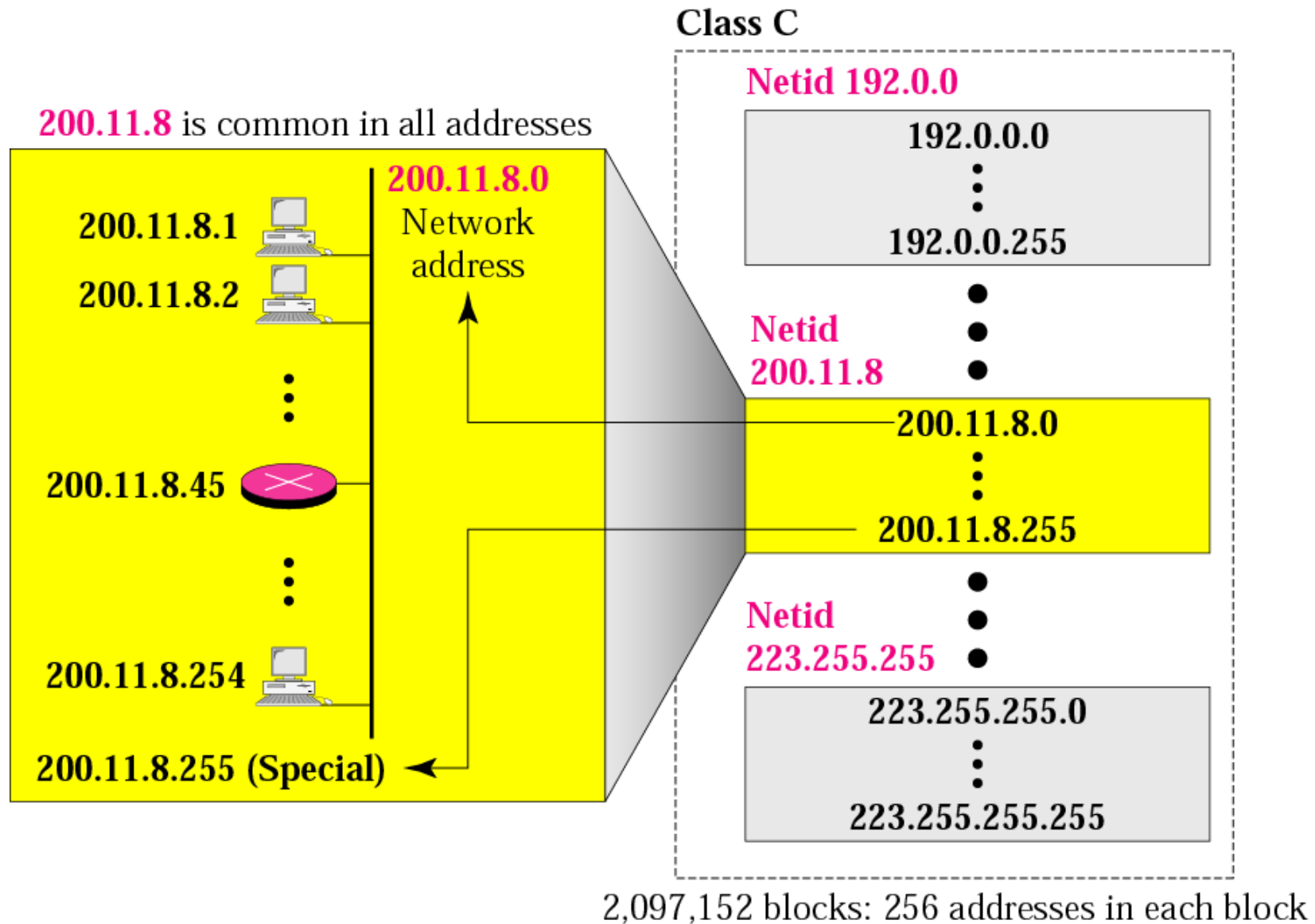
Blocks in class B[5]



16,384 blocks: 65,536 addresses in each block

Figure 4-9

Blocks in class C[5]



Summary

- Millions of *class A addresses are wasted*.
- Many *class B addresses are wasted*.
- The number of addresses in a class *C block is smaller than the needs of most organizations*.
- *Class D* addresses are used for *multicasting*; there is only one block in this class.
- *Class E* addresses are reserved for *special purposes*; most of the block is wasted.
- Network address
 - In classful addressing, the *network address* (the first address in the block) is the one that is assigned to the organization.
 - The *network address defines the network to the rest of the Internet*.
 - *Using network address*, we can find the *class of the address, the block, and the range of the addresses in the block*.

Example 1 [5]

Given the network address 17.0.0.0, find the class, the block, and the range of the addresses.

Solution

The **class is A** because the first byte is between 0 and 127. The block has a **netid of 17**. The **addresses range** from 17.0.0.0 to 17.255.255.255.

Example 2[5]

Given the network address 132.21.0.0, find the class, the block, and the range of the addresses.

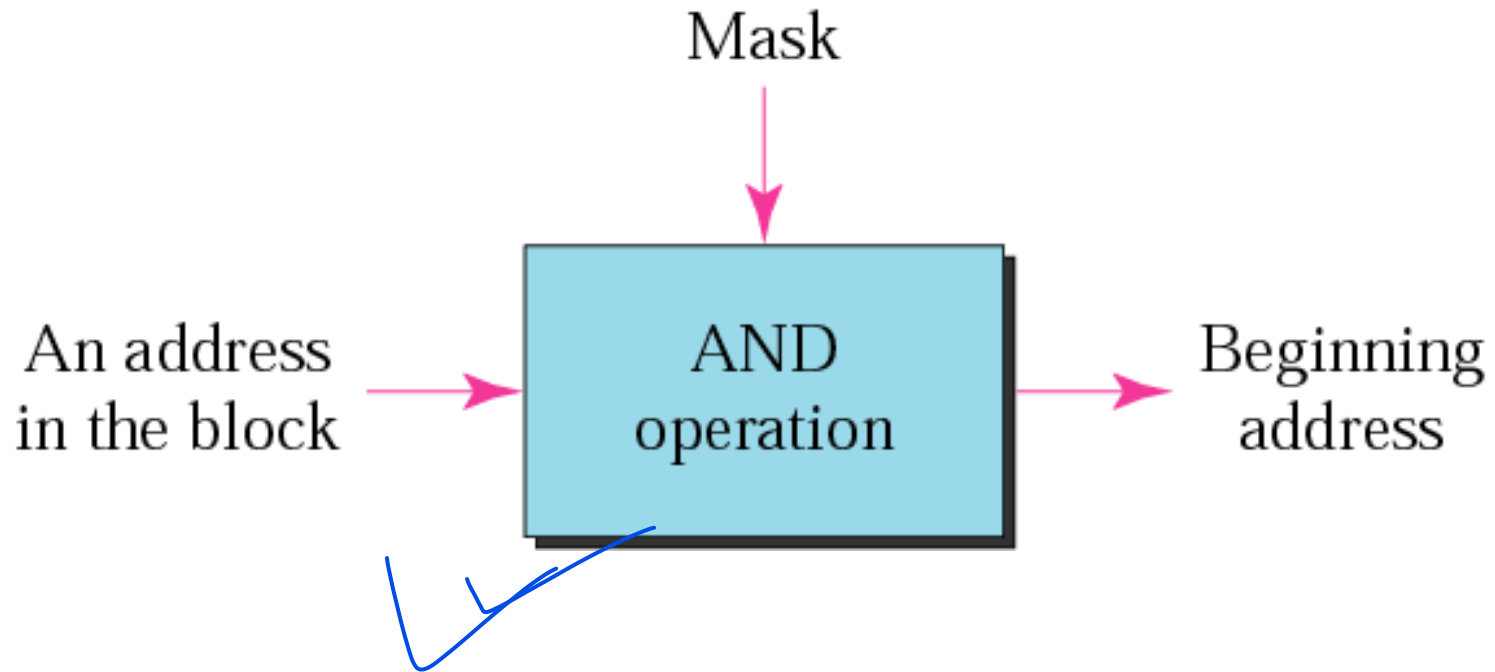
Solution

The **class is B** because the first byte is between 128 and 191. The block has a **netid of 132.21**. The **addresses range** from 132.21.0.0 to 132.21.255.255.

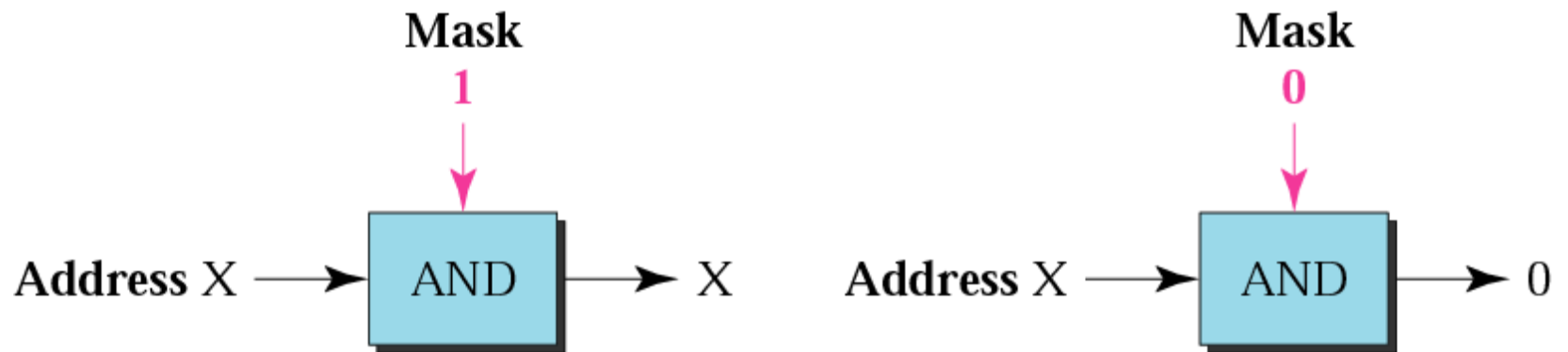
Mask

A **mask** is a 32-bit binary number that gives the first address in the block (**the network address**) when bitwise ANDed with an address in the block.

Masking concept [5]



AND operation [5]



Mask

- The **network address** is the beginning address of each block.
- It can be found by applying the **default mask** to any of the addresses in the block (including itself).
- It retains the **netid** of the block and sets the **hostid** to zero.
- *We must not apply the default mask of one class to an address belonging to another class.*

Example 4[5]

Given the address 23.56.7.91 and the default class A mask, find the beginning address (network address).

Solution

The default mask is 255.0.0.0, which means that only the first byte is preserved and the other 3 bytes are set to 0s.

The **network address** is 23.0.0.0.

Example 5[5]

Given the address 132.6.17.85 and the default class B mask, find the beginning address (network address).

Solution

The default mask is 255.255.0.0, which means that the first 2 bytes are preserved and the other 2 bytes are set to 0s.

The **network address** is 132.6.0.0.

Private Addresses

A number of blocks in each class are assigned for private use. They are not recognized globally. These blocks are depicted as follows:

Class A: 10.0.0(Net id) and no of Blocks is 1

7.

Class B: 172.16-172.31 and 16

Class C: 192.168.0-192.168.255 and 256 blocks

Unicast, Multicast, and Broadcast (Casting)

Unicast communication is *one-to-one*.

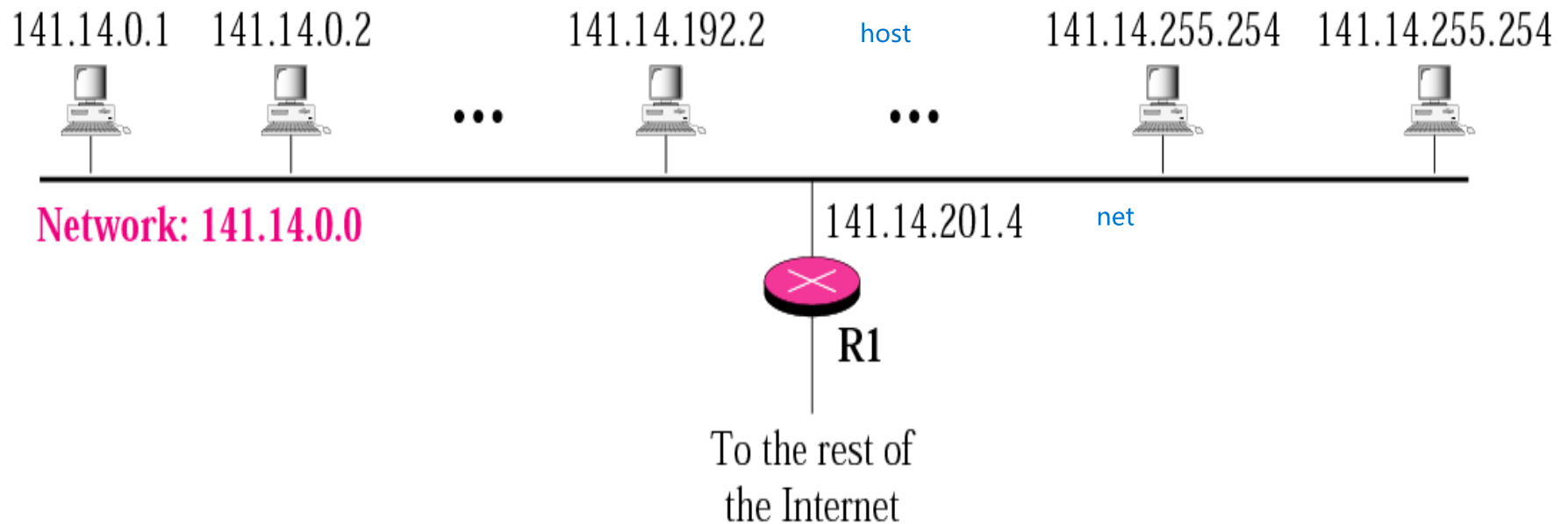
Multicast communication is *one-to-many*.

Broadcast communication is *one-to-all*.

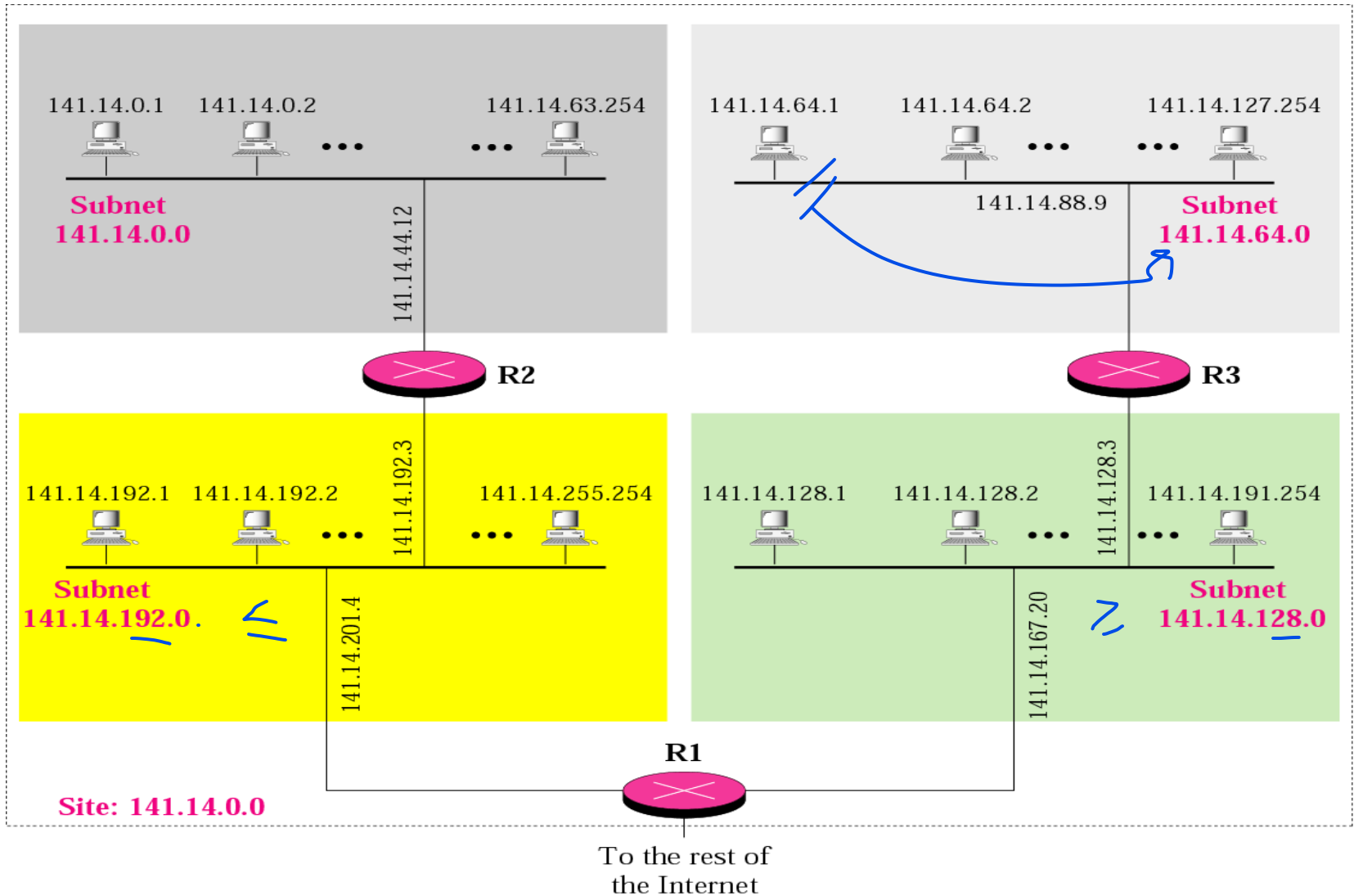
Note

IP addresses are designed with two levels of hierarchy.

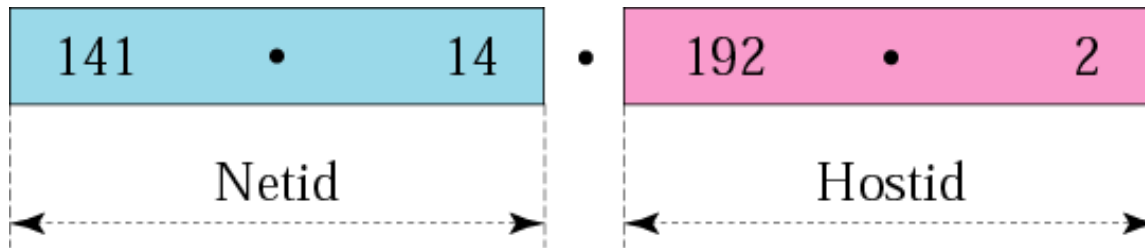
A network with two levels of hierarchy (not subnetted) [5]



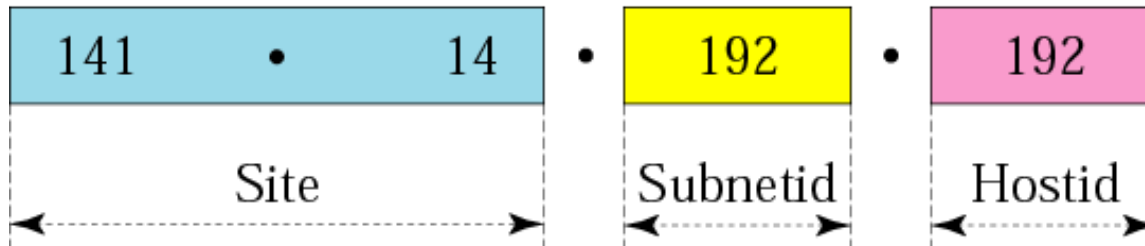
A network with three levels of hierarchy (sub netted) [5]



Addresses in a network with and without sub netting [5]



a. Without subnetting

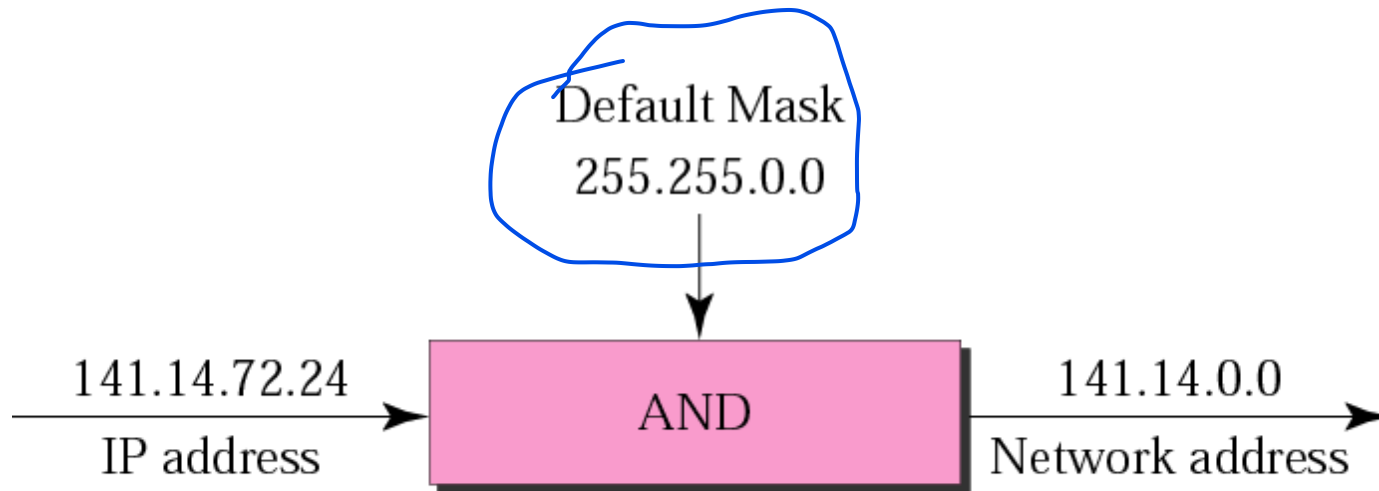


b. With subnetting

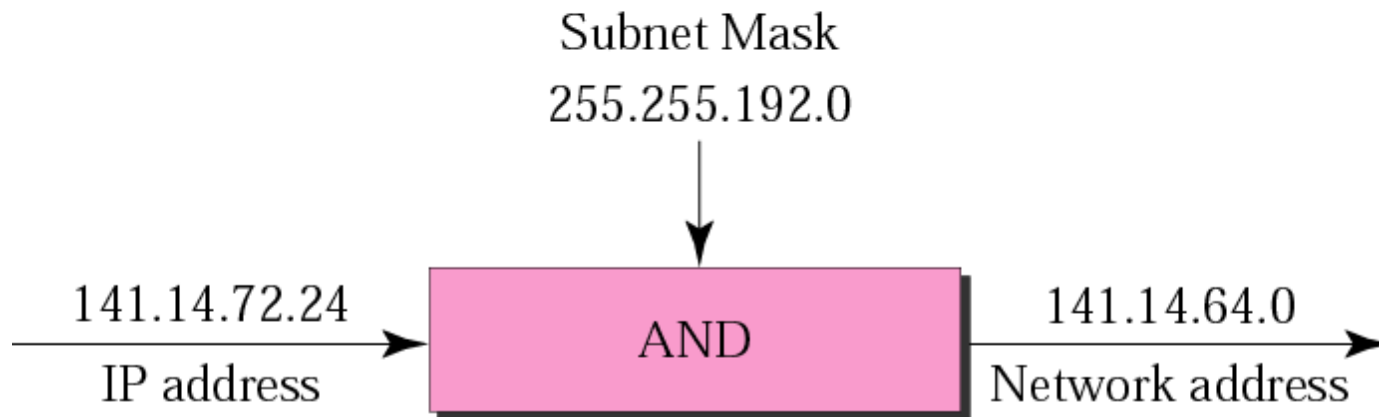
Hierarchy concept in a telephone number

(408) 864 – 8902
Area code Exchange Connection

Default mask and subnet mask [5]



a. Without subnetting



b. With subnetting

Finding the Subnet Address

Given an IP address, we can find the subnet address the same way we found the network address in the previous slide. We apply the mask to the address.

Short-Cut Method [5]

- *** If the byte in the mask is 255, copy the byte in the address.
- *** If the byte in the mask is 0, replace the byte in the address with 0.
- *** If the byte in the mask is neither 255 nor 0, we write the mask and the address in binary and apply the AND operation.

Example 8[5]

What is the subnetwork address if the destination address is 19.30.84.5 and the mask is 255.255.192.0?

Solution

See Figure 1

Example 8 [5]

IP Address

19	•	30	•	84	•	5
----	---	----	---	----	---	---

Mask

255	•	255	•	192	•	0
-----	---	-----	---	-----	---	---

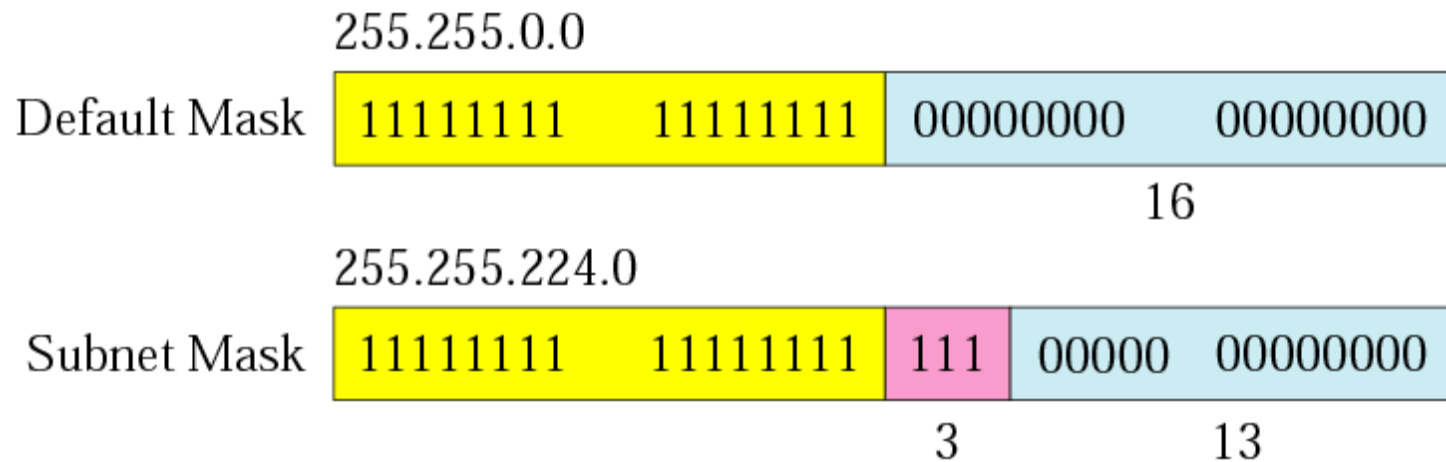
19	•	30	•	64	•	0
----	---	----	---	----	---	---

Subnet Address

↓

84	0	1	0	1	0	1	0	0
192	1	1	0	0	0	0	0	0
64	0	1	0	0	0	0	0	0

Comparison of a default mask and a subnet mask [5]



Note

The number of subnets must be a power of 2.

Example 9[5]

A company is granted the site address 201.70.64.0 (class C). The company needs six subnets. Design the subnets.

Solution

The number of 1s in the default mask is 24 (class C).

Solution (Continued)

The company needs six subnets. This number 6 is not a power of 2. The next number that is a power of 2 is 8 (2^3). We need 3 more 1s in the subnet mask. The total number of 1s in the subnet mask is 27 ($24 + 3$).

The total number of 0s is 5 ($32 - 27$). The mask is

Solution (Continued)

11111111 11111111 11111111 11100000

or

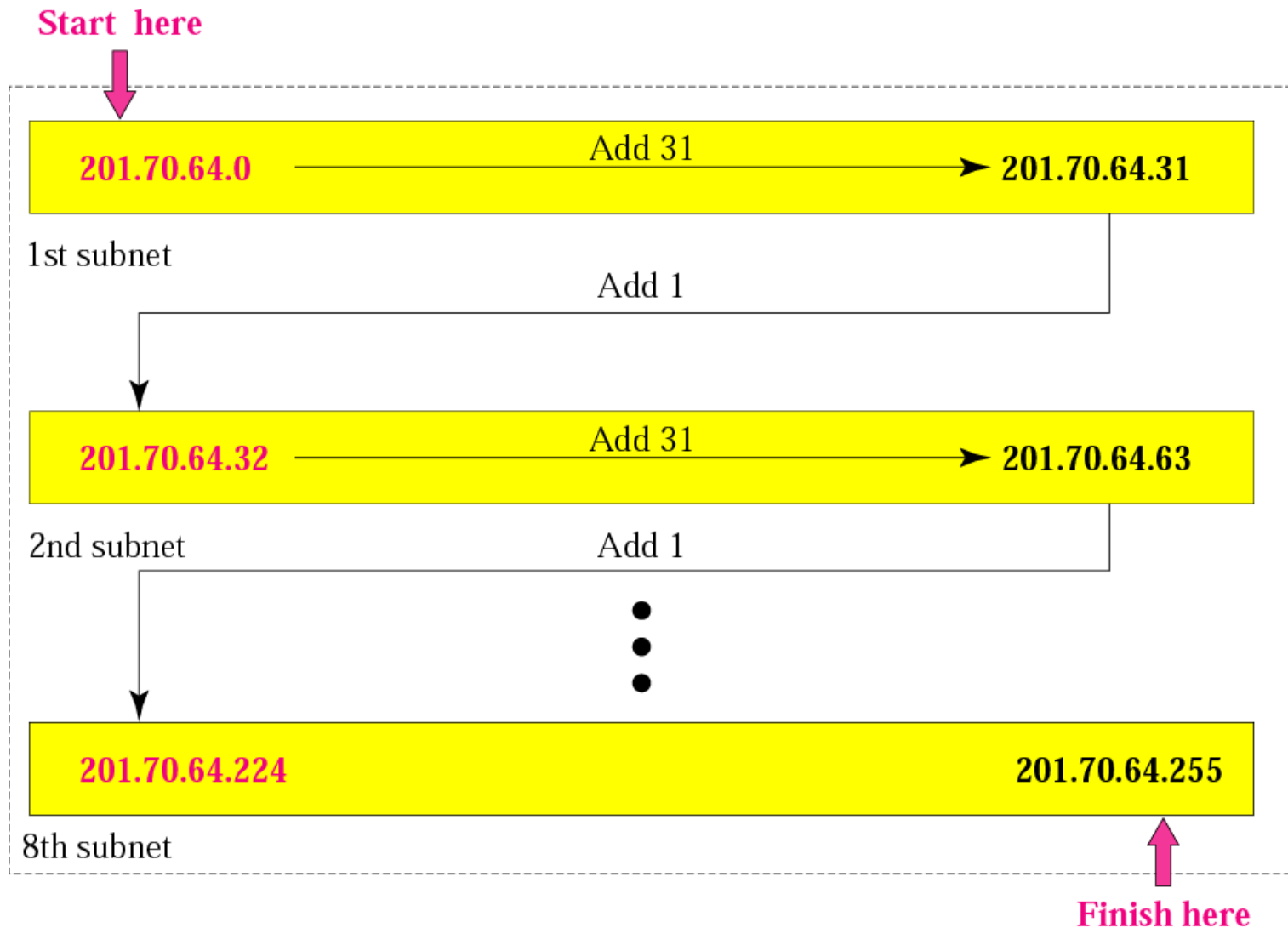
255.255.255.224

The number of subnets is 8.

The number of addresses in each subnet is 2^5 (5 is the number of 0s) or 32.

See Figure 5.8

Example 9[5]



Example 10[5]

A company is granted the site address 181.56.0.0 (class B). The company needs 1000 subnets. Design the subnets.

Solution

The number of 1s in the default mask is 16 (class B).

Solution (Continued)

The company needs 1000 subnets. This number is not a power of 2. The next number that is a power of 2 is 1024 (2^{10}). We need 10 more 1s in the subnet mask.

The total number of 1s in the subnet mask is 26 ($16 + 10$).

The total number of 0s is 6 ($32 - 26$).

Solution (Continued)

The mask is

11111111 11111111 11111111 11000000

or

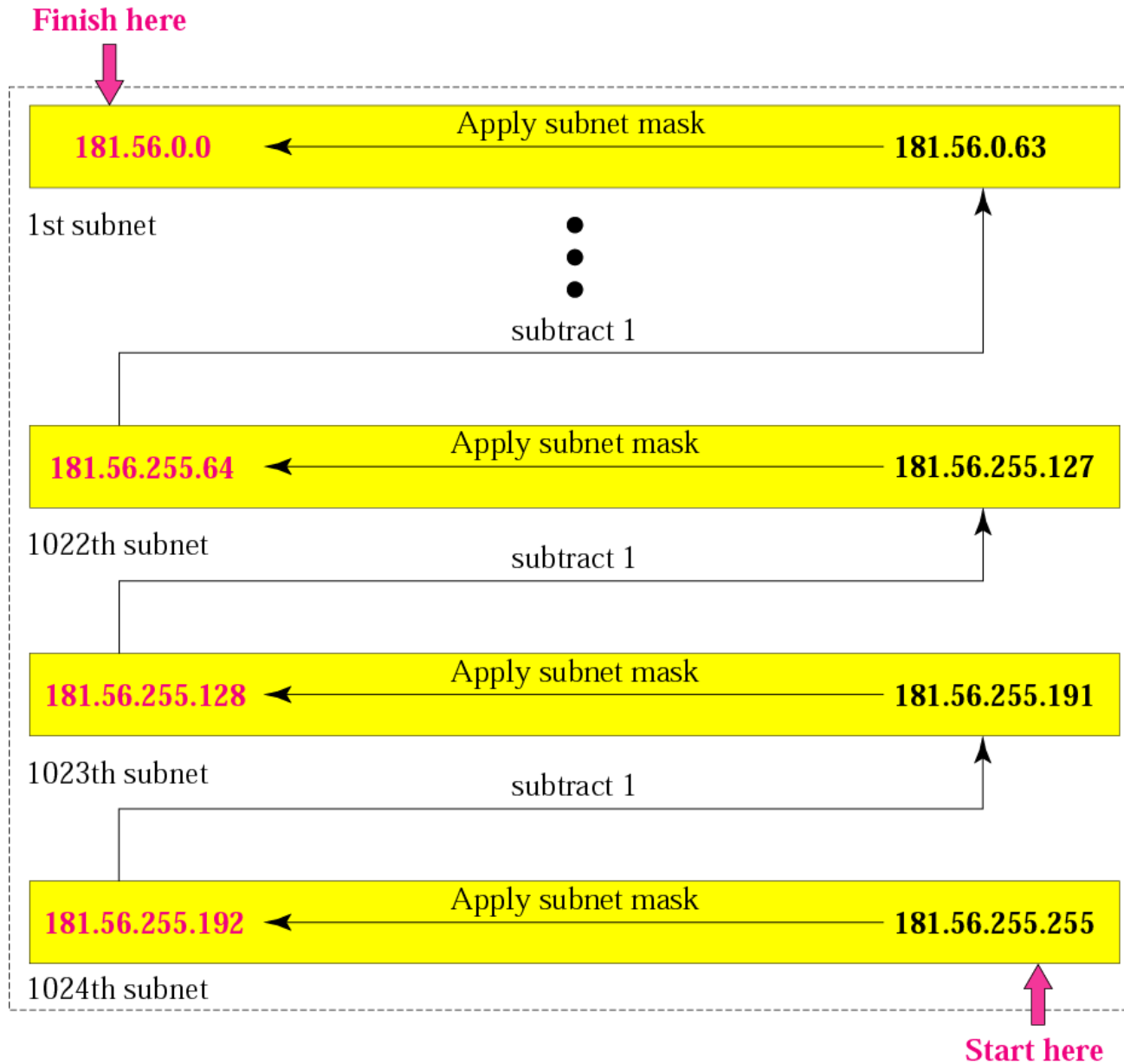
255.255.255.192.

The number of subnets is 1024.

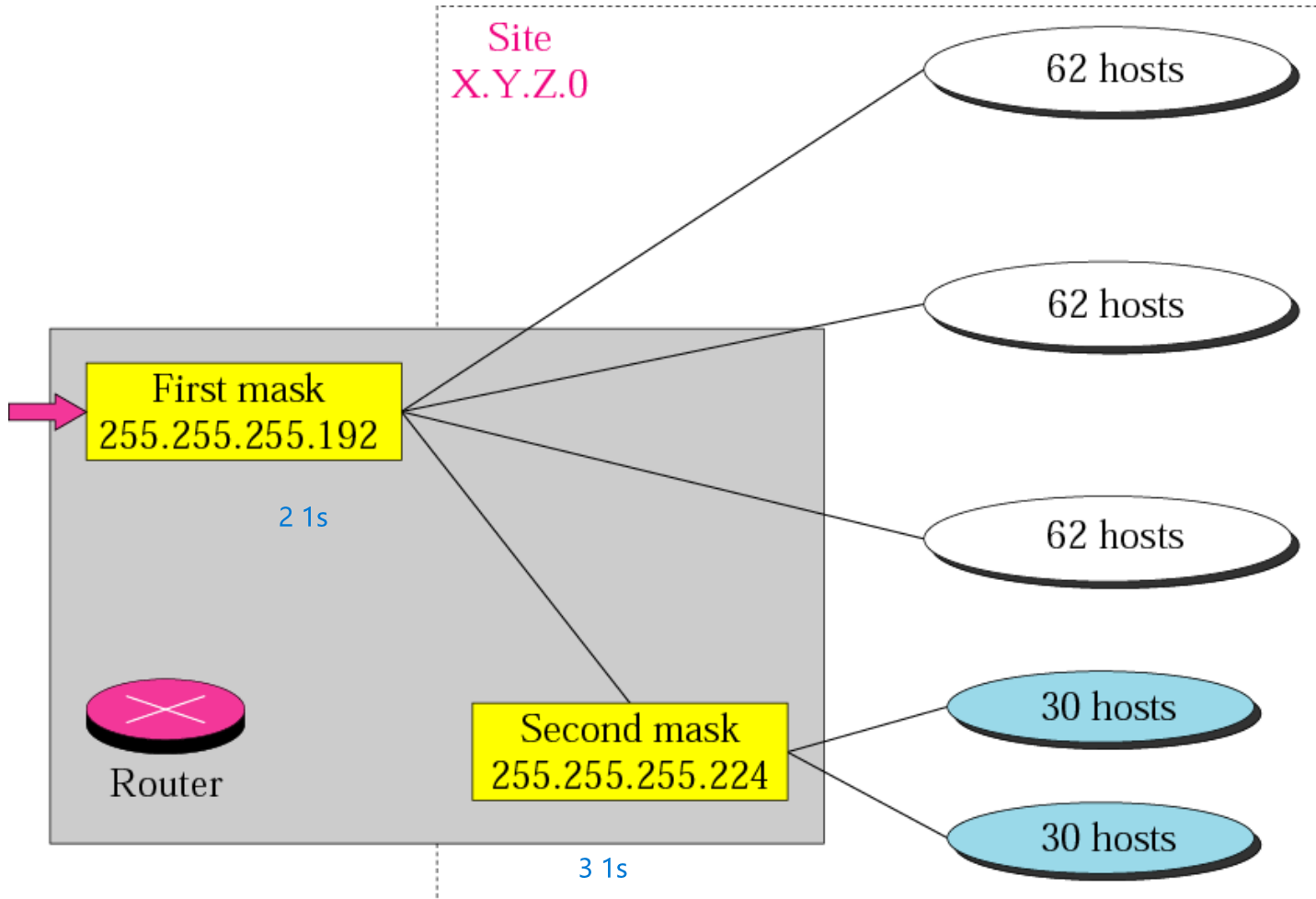
The number of addresses in each subnet is 2^6
(6 is the number of 0s) or 64.

See Figure 2

Example 10[5]



Variable-length sub netting [5]



Q Consider the network having IP Address 200.1.2.0. Divide this network into two subnets. i

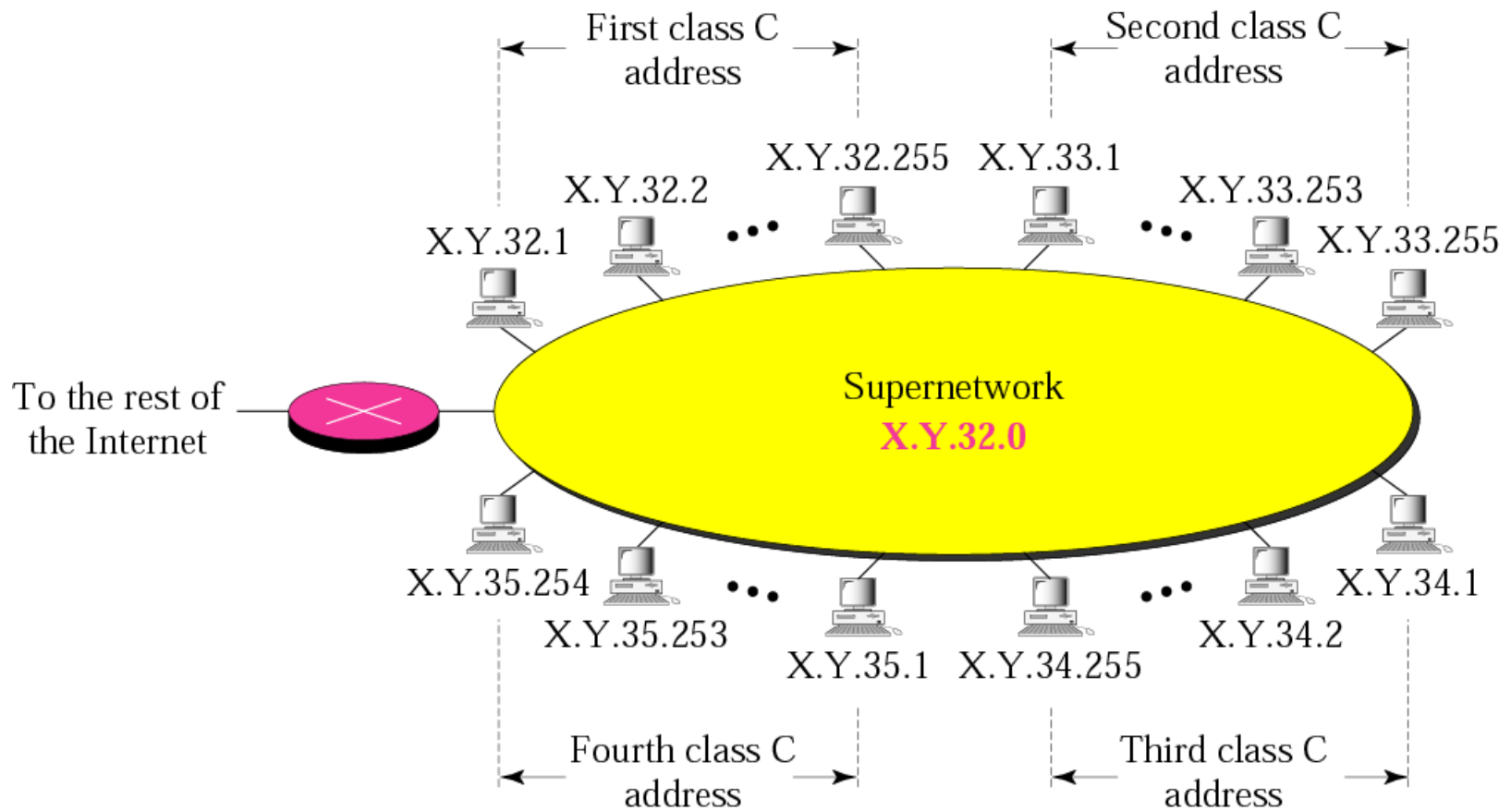
1st Subnet

- IP Address of the subnet / Subnet id = 200.1.2.0
- Direct Broadcast Address = $200.1.2.01111111 = 200.1.2.127$
- Total number of IP Addresses = $2^7 = 128$
- Range of IP Addresses = [200.1.2.0, 200.1.2.127]
- Total number of hosts that can be configured = $128 - 2 = 126$
- Range of Allocated IP Addresses = [200.1.2.1, 200.1.2.126]

2nd Subnet

- IP Address of the subnet / Subnet id = 200.1.2.128
- Direct Broadcast Address = $200.1.2.11111111 = 200.1.2.255$
- Total number of IP Addresses = $2^7 = 128$
- Range of IP Addresses = [200.1.2.128, 200.1.2.255]
- Total number of hosts that can be configured = $128 - 2 = 126$
- Range of Allocated IP Addresses = [200.1.2.129, 200.1.2.254]

A super network [5]



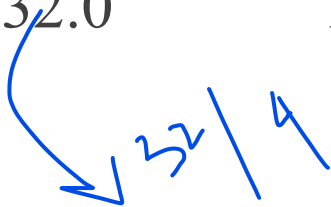
Rules:

- *** The number of blocks must be a power of 2 (1, 2, 4, 8, 16, . . .).
- *** The blocks must be contiguous in the address space (no gaps between the blocks).
- *** The third byte of the first address in the superblock must be evenly divisible by the number of blocks. In other words, if the number of blocks is N , the third byte must be divisible by N .

Example 11[5]

A company needs 600 addresses. Which of the following set of class C blocks can be used to form a super net for this company?

1. 198.47.32.0 198.47.33.0 198.47.34.0
2. 198.47.32.0 198.47.42.0 198.47.52.0 198.47.62.0
3. 198.47.31.0 198.47.32.0 198.47.33.0 198.47.52.0
4. 198.47.32.0 198.47.33.0 198.47.34.0 198.47.35.0



Solution

- 1: No, there are only three blocks.
- 2: No, the blocks are not contiguous.
- 3: No, 31 in the first block is not divisible by 4.
- 4: Yes, all three requirements are fulfilled.

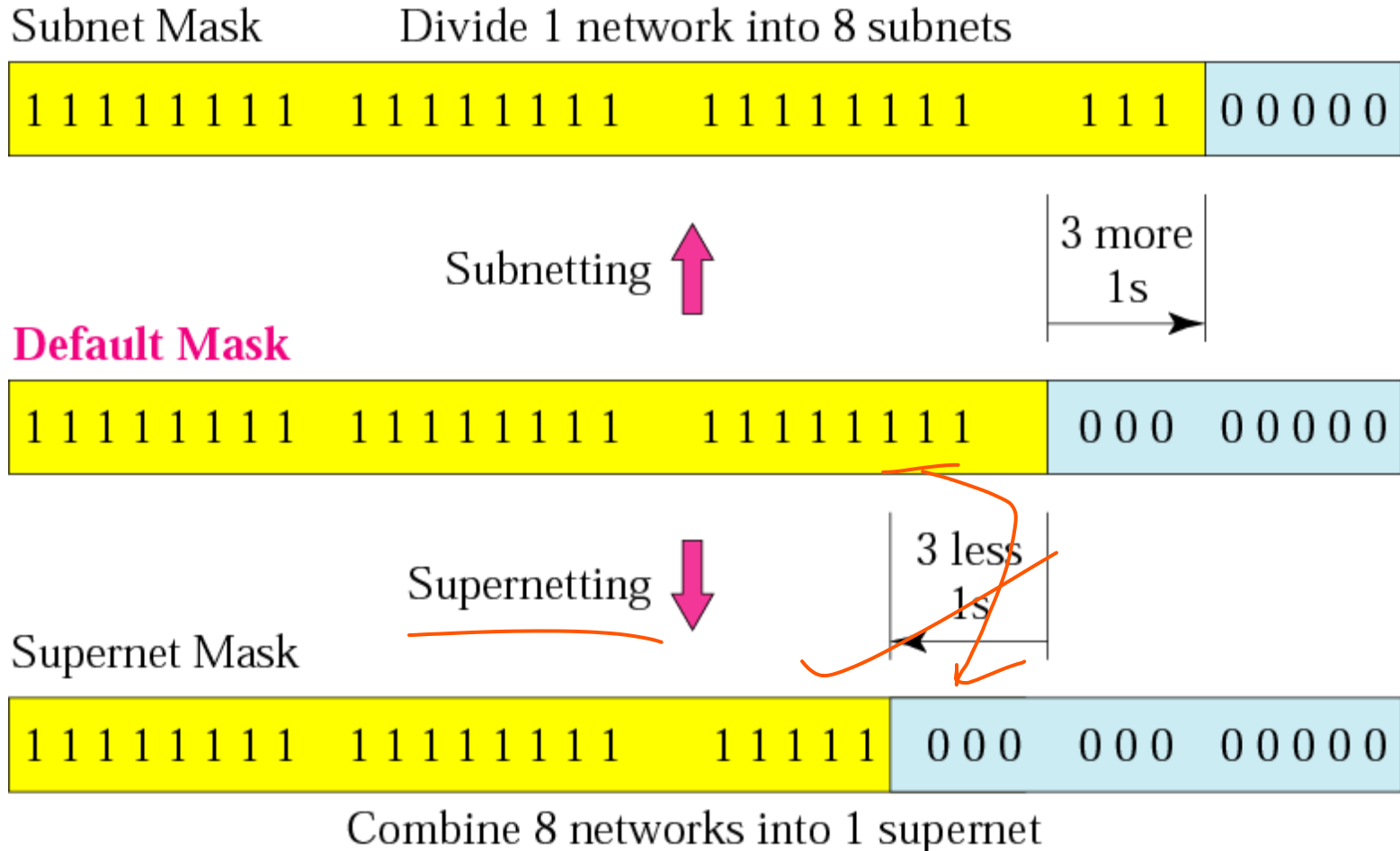
Note

*In subnetting,
we need the first address of the
subnet and the subnet mask to
define the range of addresses.*

Note

In supernetting,
we need the first address of
the supernet
and the supernet mask to
define the range of addresses.

Comparison of subnet, default, and supernet masks [5]



Example 12[5]

We need to make a super-network out of 16 class C blocks. What is the supernet mask?

Solution

We need 16 blocks. For 16 blocks we need to change four 1s to 0s in the default mask. So the mask is

11111111 11111111 11110000 00000000

or

255.255.240.0

Example 13[5]

A supernet has a first address of 205.16.32.0 and a supernet mask of 255.255.248.0. A router receives three packets with the following destination addresses:

205.16.37.44

205.16.42.56

205.17.33.76



Which packet belongs to the supernet?

Solution

We apply the supernet mask to see if we can find the beginning address.

205.16.37.44 AND 255.255.248.0 → 205.16.32.0

205.16.42.56 AND 255.255.248.0 → 205.16.40.0

205.17.33.76 AND 255.255.248.0 → 205.17.32.0

Only the first address belongs to this supernet.

Classless Addressing

- Is it possible to assign any number of network id or host id to any user in classful addressing?
 - No.
 - Classes are well defined
 - Network ID bits and Host ID bits are fixed.
- Solution is **Classless Addressing**

Classless Addressing

- It is also known by **CIDR** (Classless Inter Domain Routing) notation.
- Slash notation is also called CIDR
- It is denoted as **A.B.C.D/n**
- **n represents the number of bits in network ID**
- Any number of host can be assigned to any user.

Example 15[5]

A small organization is given a block with the beginning address and the prefix length 205.16.37.24/29 (in slash notation). What is the range of the block?

Solution

The beginning address is 205.16.37.24. To find the last address we keep the first 29 bits and change the last 3 bits to 1s.

Beginning: 11001111 00010000 00100101 00011000

Ending : 11001111 00010000 00100101 00011111

There are only 8 addresses in this block.

Solution (contd.)

We can find the range of addresses in Example 15 by another method. We can argue that the length of the suffix is $32 - 29$ or 3. So there are $2^3 = 8$ addresses in this block. If the first address is 205.16.37.24, the last address is 205.16.37.31 ($24 + 7 = 31$).

1 w/p

Note

A block in classes A, B, and C
can easily be represented in slash
notation as

A.B.C.D/ n

where n is
either 8 (class A), 16 (class B), or
24 (class C).

Example 16[5]

What is the network address if one of the addresses is 167.199.170.82/27?

Solution

The prefix length is 27, which means that we must keep the first 27 bits as is and change the remaining bits (5) to 0s. The 5 bits affect only the last byte. The last byte is 01010010. Changing the last 5 bits to 0s, we get 01000000 or 64. The network address is 167.199.170.64/27.



Example 17[5]

An organization is granted the block 130.34.12.64/26. The organization needs to have four subnets. What are the subnet addresses and the range of addresses for each subnet?

Solution

The suffix length is 6. This means the total number of addresses in the block is 64 (2^6). If we create four subnets, each subnet will have 16 addresses.

Solution (Continued)

Let us first find the subnet prefix (subnet mask). We need four subnets, which means we need to add two more 1s to the site prefix. The subnet prefix is then /28.

Subnet 1: 130.34.12.64/28 to 130.34.12.79/28.

Subnet 2 : 130.34.12.80/28 to 130.34.12.95/28.

Subnet 3: 130.34.12.96/28 to 130.34.12.111/28.

Subnet 4: 130.34.12.112/28 to 130.34.12.127/28.

(Covert these numbers in binary and then change the last 4 Bits to see the actual addressing)

Example 18[5]

Variable Length Subnet Masking in CIDR

An ISP is granted a block of addresses starting with 190.100.0.0/16. The ISP needs to distribute these addresses to three groups of customers as follows:

1. The first group has 64 customers; each needs 256 addresses.
 2. The second group has 128 customers; each needs 128 addresses.
 3. The third group has 128 customers; each needs 64 addresses.
- Design the subblocks and give the slash notation for each subblock. Find out how many addresses are still available after these allocations.

Solution

Group 1

For this group, each customer needs 256 addresses. This means the suffix length is 8 ($2^8 = 256$). The prefix length is then $32 - 8 = 24$.

01: 190.100.0.0/24 → 190.100.0.255/24

02: 190.100.1.0/24 → 190.100.1.255/24

.....

64: 190.100.63.0/24 → 190.100.63.255/24

Total = $64 \times 256 = 16,384$

Solution (Continued)

Group 2

For this group, each customer needs 128 addresses. This means the suffix length is 7 ($2^7 = 128$). The prefix length is then $32 - 7 = 25$. The addresses are:

001: 190.100.64.0/25 ➔ 190.100.64.127/25

002: 190.100.64.128/25 ➔ 190.100.64.255/25

.....

128: 190.100.127.128/25 ➔ 190.100.127.255/25

Total = $128 \times 128 = 16,384$

Solution (Continued)

Group 3

For this group, each customer needs 64 addresses. This means the suffix length is 6 ($2^6 = 64$). The prefix length is then $32 - 6 = 26$. The addresses are:

001: 190.100.128.0/26 → 190.100.128.63/26

002: 190.100.128.64/26 → 190.100.128.127/26

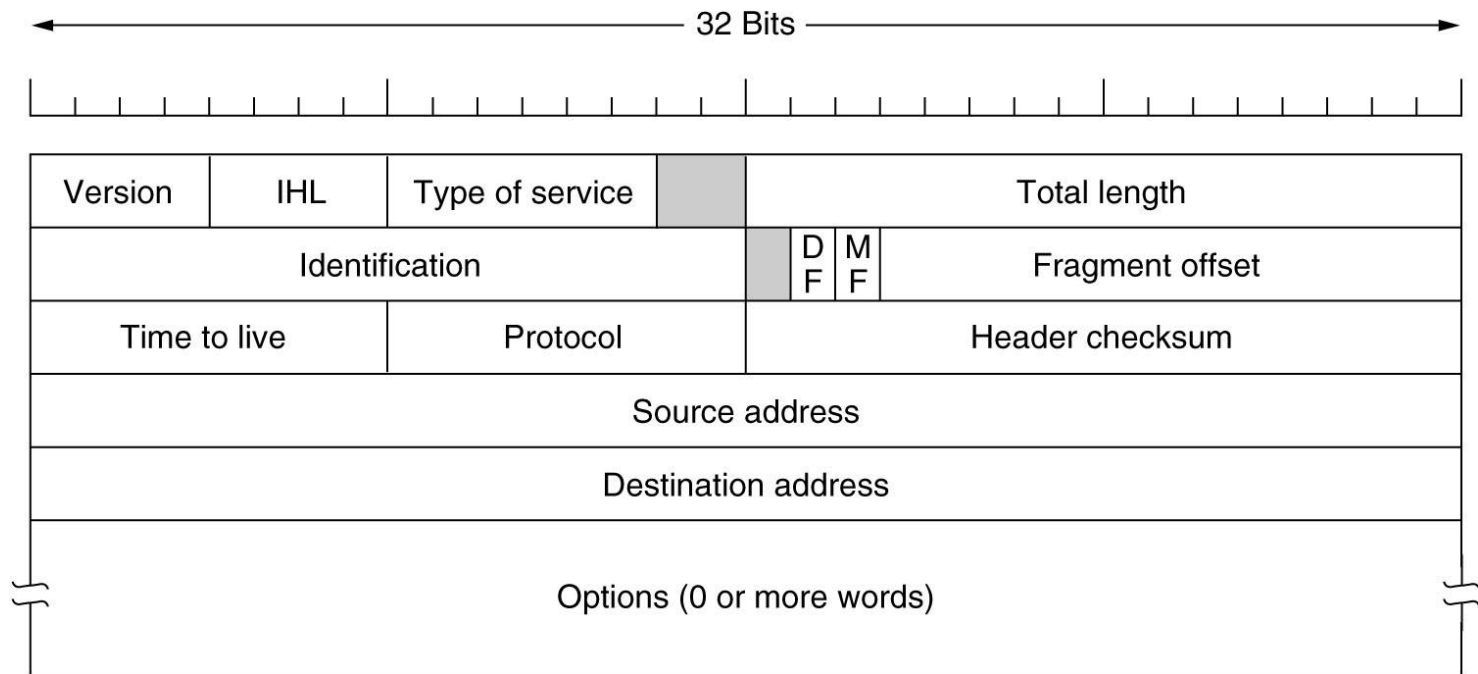
.....

128: 190.100.159.192/26 → 190.100.159.255/26

Total = $128 \times 64 = 8,192$

IP Protocol [2]

- Internet Protocol (IP) is the glue
 - It facilitates packets to be transported across different types of networks, from source host to destination host



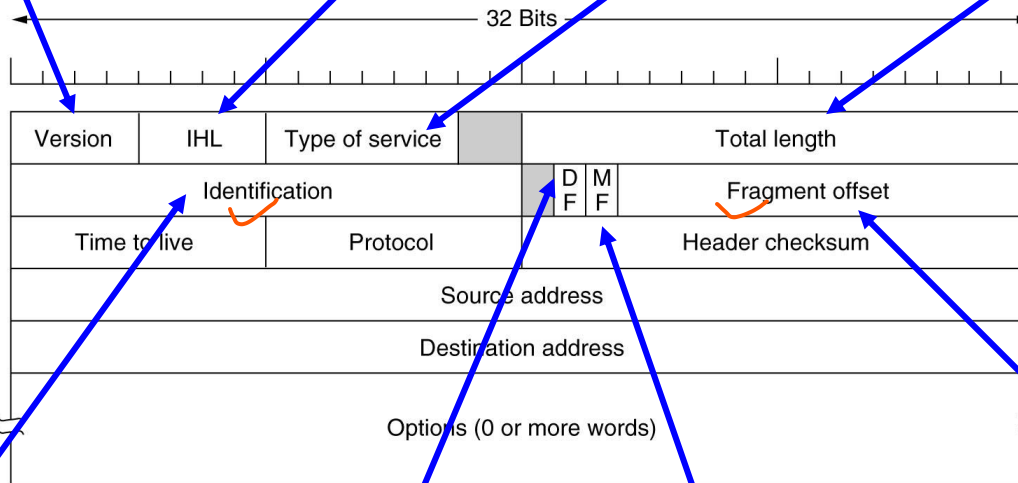
IP packet format [2]

version of the IP protocol (4bits)

IP header length (4bits) multiply to 4 in Bytes.

used for DiffServ 8 bits (DSCP), D,T,R

length of header + payload, 16 bits = 2^{16} is datagram



vhtl
if
tph
s
d
o

Unique packet id 16 bits (to combine packets)

“do not fragment”

“more fragments”

Total 3 bits

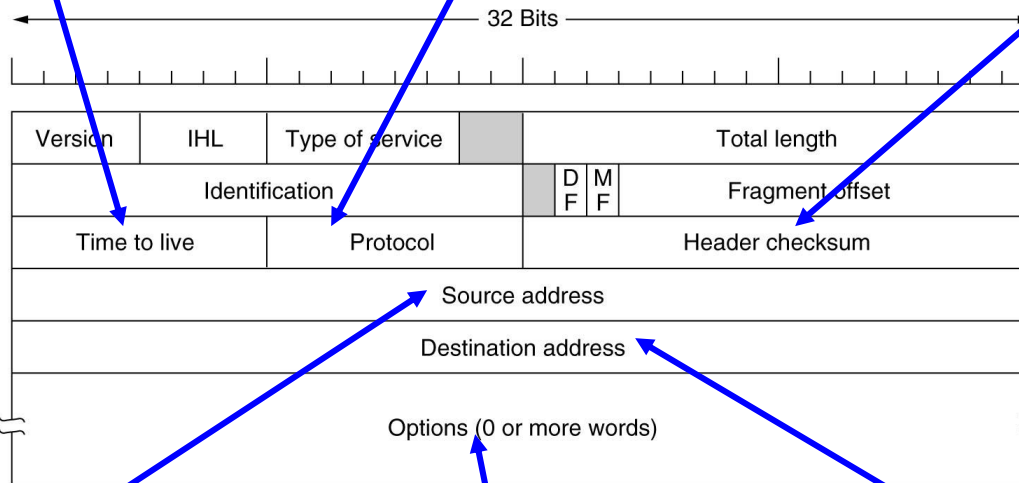
Position of Fragment in IP Packets 13 bits

IP Packet Format [2]

Helps to limit the no. of hops or time spent in the network (set initially)

Protocol used to generate the payload (ICMP-1, TCP-6, UDP-17 etc.)

16 bit checksum, check if packet is error free

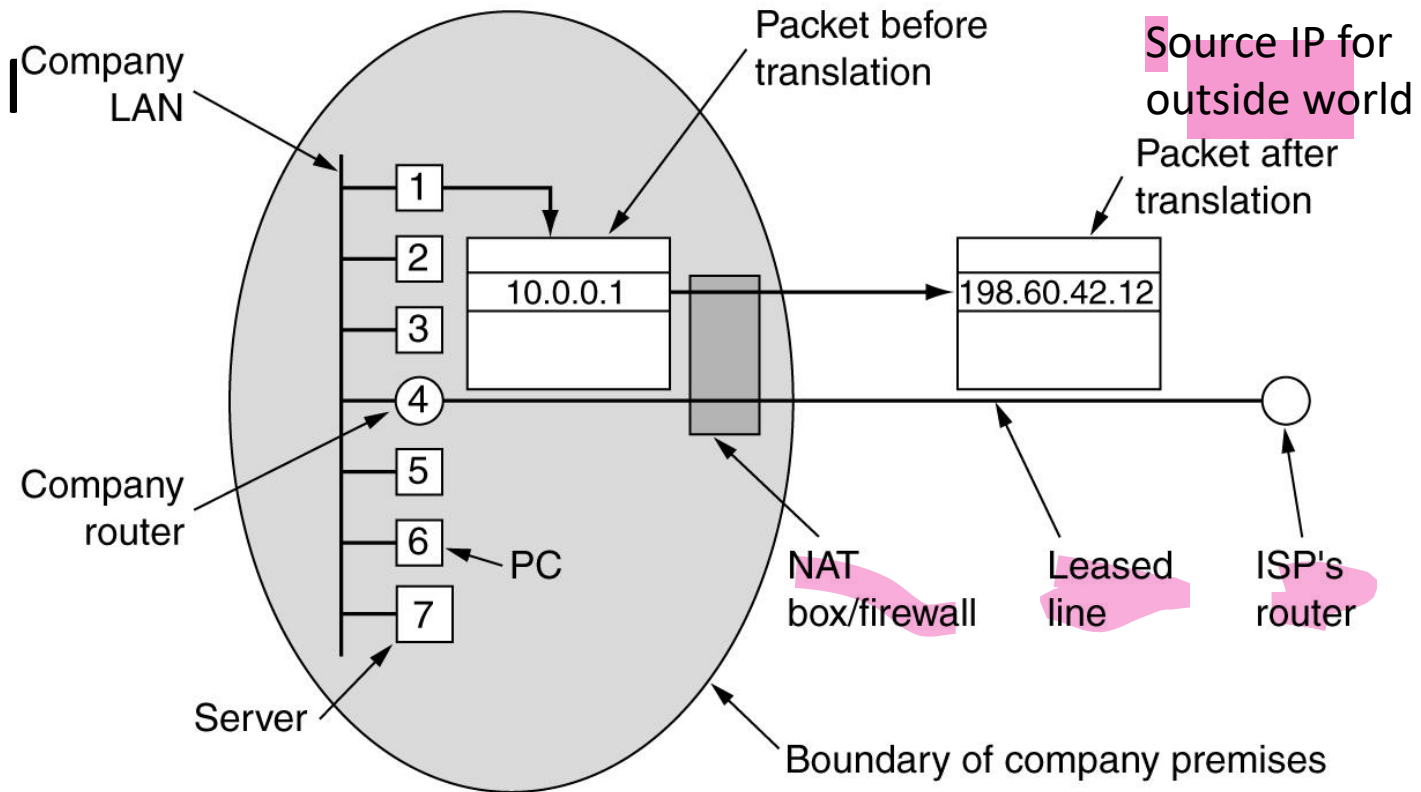


Source IP address

Optional information, such as security, record route and Time Stamp

Destination IP address

NAT – Network Address Translation [2]



IPv6 [6]

- **Initial motivation:** 32-bit address space soon to be completely allocated.
 - IPv4 address is **128 bits**. 2^{96} more than IPv4.
 - One per ms – 1000 address every second – **15 Billion year later not even 1%** addresses will be allocated.
 - Additional motivation:
 - header format helps speed processing/forwarding
 - header changes to facilitate QoS
- IPv6 datagram format:**
- fixed-length 40 byte header (IPv4 – 20 to 60)
 - no fragmentation allowed

IPv6 Header (Cont) [6]

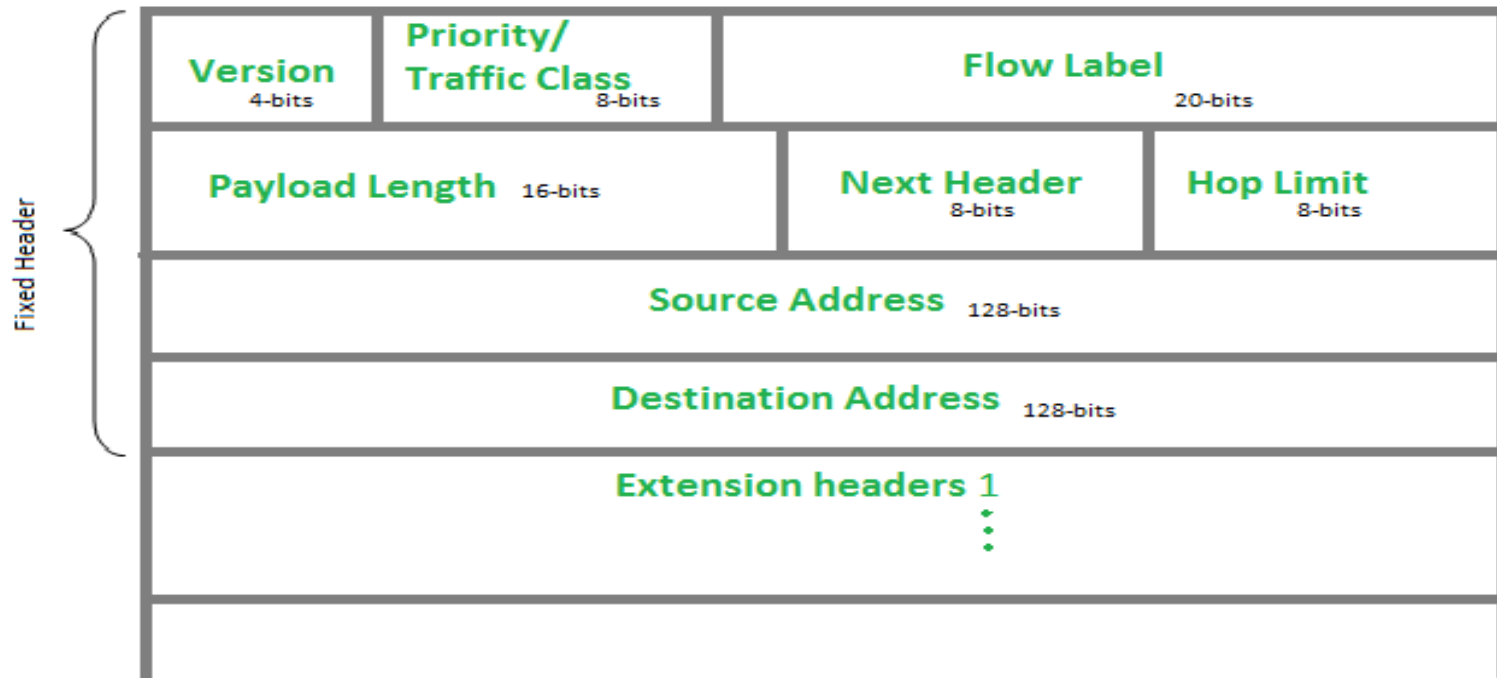
Priority: identify priority among datagrams in flow. Priority can be set to overcome congestion. **Least priority** packets are **dropped**.

Flow Label: identify datagrams in same “flow.” **Works like TCP.**

Direction of the packet flow is set using the virtual circuit.
(concept of “flow” not well defined).

Payload: Data length in Bytes (65535)

vpf
pnh
s
d



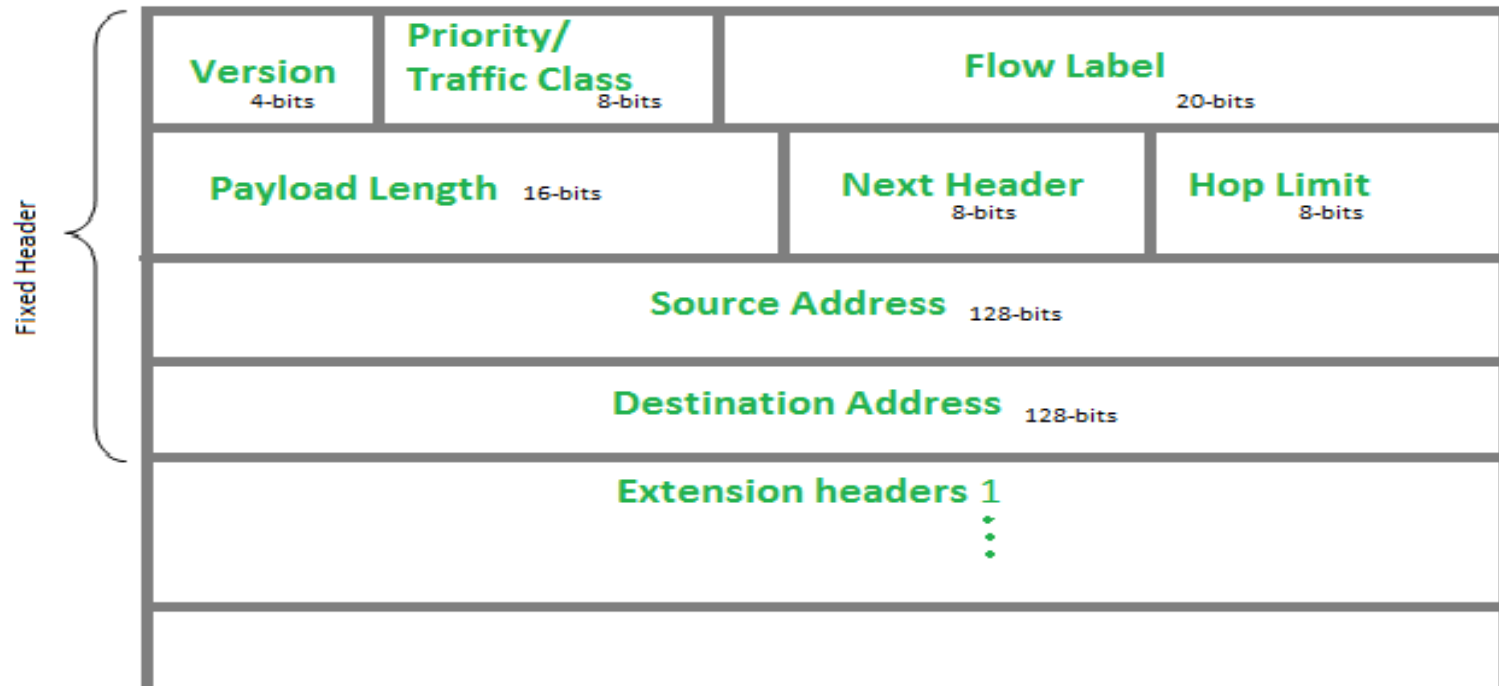
IPv6 Header (Cont) [6]

Next header: Indicates type of *extension header*. Some cases to identify the upper layer protocol for data such as TCP, UDP etc.

Hop Count: Same as TTL in IPv4. Max hops a packet can travel.

Source address: 128bits original source of packet.

Destination address: 128bits final destination address.



Other Changes from IPv4

COI

- *Checksum*: removed entirely to reduce processing time at each hop
- *Options*: allowed, but outside of header,
- *ICMPv6*: new version of ICMP
 - additional message types, e.g. “Packet Too Big”
 - multicast group management functions

Transition From IPv4 To IPv6

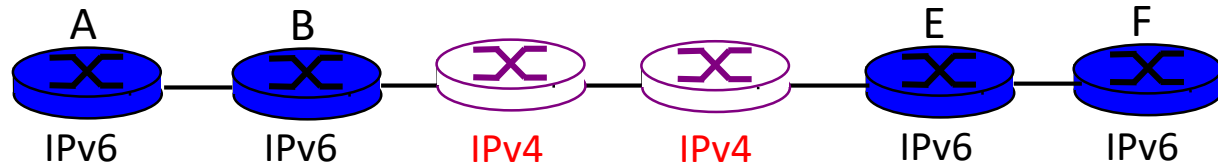
- Not all routers can be upgraded simultaneous
 - no “flag days”
 - How will the network operate with mixed IPv4 and IPv6 routers?
- *Dual Stack:* each node will have IPv4/IPv6 datagram (for detail refer Kurose book)
- *Tunneling:* IPv6 carried as payload in IPv4 datagram among IPv4 routers

Tunneling

Logical view:



Physical view:

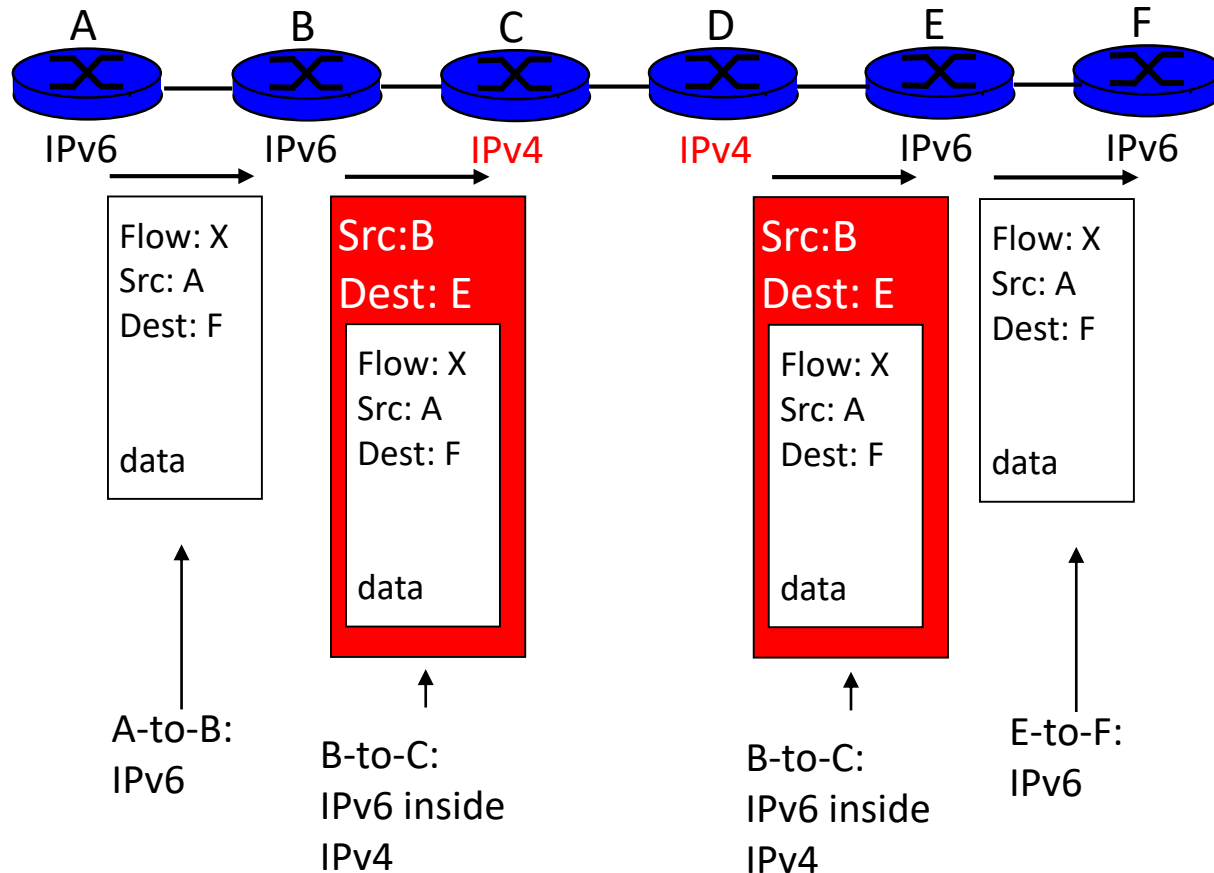


Tunneling [6]

Logical view:

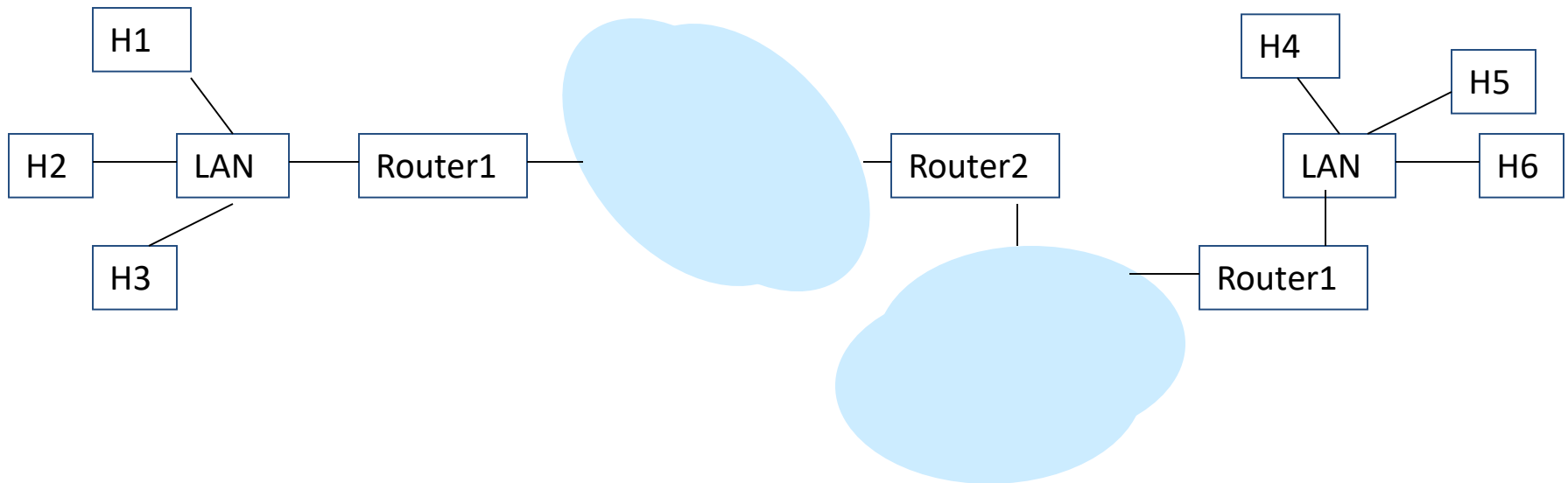


Physical view:



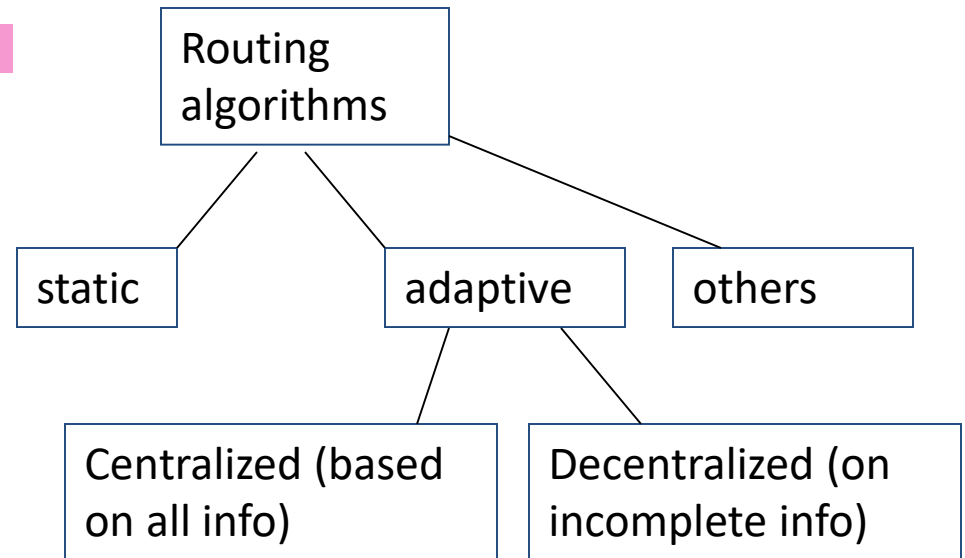
Routing: The problem

- Largely concerned with routing datagram's through a subnet
- Between a pair of source-destination devices, packets may have to traverse several “subnets”
- Routing tables are updated every T seconds



Routing protocols: classification

- Static routes
 - Computed off-line
 - based on certain topology, traffic, performance metric
 - Not change, unless there is a major network overhaul
- Adaptive routing
 - Routes adapt to changes in topology, traffic
 - On-line based on current measurements
 - Based on complete or partial knowledge
 - Distributed computation vs. centralized computation
- Other algorithms
 - Flooding
 - Broadcasting



Shortest Path Routing Algorithm [6]

Dijkstra's algorithm

- net topology, link costs known to all nodes
 - accomplished via “link state broadcast”
 - all nodes have same info
- computes least cost paths from one node (“source”) to all other nodes
 - gives forwarding table for that node
- iterative: after k iterations, know least cost path to k dest.'s

Notation:

- $c(x,y)$: link cost from node x to y ; $= \infty$ if not direct neighbors
- $D(v)$: current value of cost of path from source to dest. v
- $p(v)$: predecessor node along path from source to v
- N' : set of nodes whose least cost path definitively known

Dijkstra's Algorithm [6]

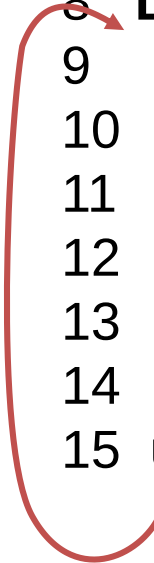
1 Initialization:

```
2  N' = {u}
3  for all nodes v
4    if v adjacent to u
5      then  $D(v) = c(u,v)$ 
6    else  $D(v) = \infty$ 
```

7

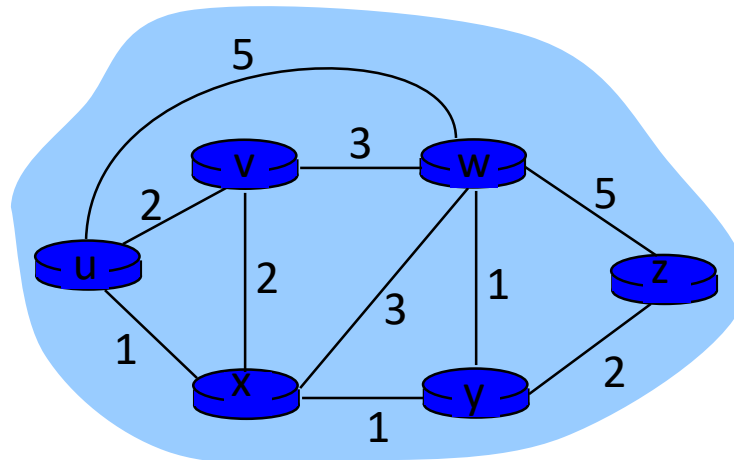
8 Loop

```
9  find w not in N' such that D(w) is a minimum
10 add w to N'
11 update D(v) for all v adjacent to w and not in N' :
12    $D(v) = \min( D(v), D(w) + c(w,v) )$ 
13 /* new cost to v is either old cost to v or known
14    shortest path cost to w plus cost from w to v */
15 until all nodes in N'
```



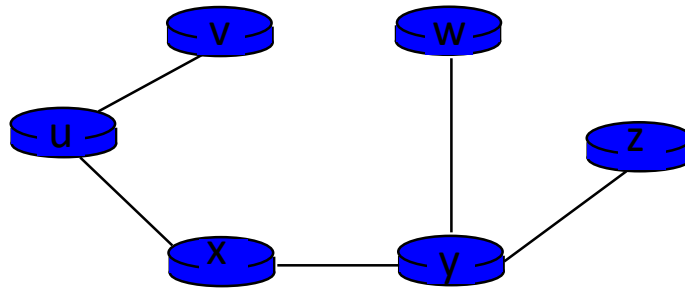
Shortest Path Routing (Dijkstra's Algorithm) [6]

Step	N'	D(v),p(v)	D(w),p(w)	D(x),p(x)	D(y),p(y)	D(z),p(z)
0	u	2,u	5,u	1,u	∞	∞
1	ux	2,u	4,x		2,x	∞
2	uxy	2,u	3,y			4,y
3	uxyv		3,y			4,y
4	uxyvw					4,y
5	uxyvwz					



Dijkstra's Algorithm(2)

Resulting shortest-path tree from u:



Resulting forwarding table in u:

destination	link
v	(u,v)
x	(u,x)
y	(u,x)
w	(u,x)
z	(u,x)

Adaptive routing

- Distance-vector routing
- Link-state routing
- Others
 - Hierarchical routing
- Standards
 - Open Shortest Path First (OSPF)
 - Border Gateway Protocol (BGP)

Distance Vector Algorithm (1) [6]

Bellman-Ford Equation (dynamic programming)

Define

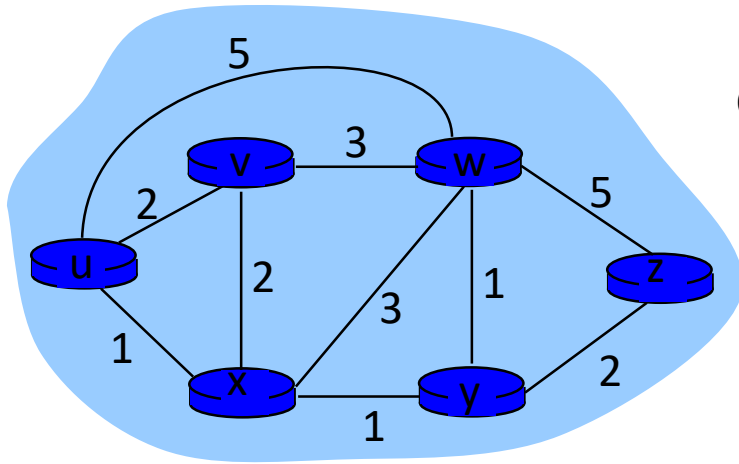
$d_x(y) :=$ cost of least-cost path from x to y

Then

$$d_x(y) = \min \{ c(x,v) + d_v(y) \}$$

where *min* is taken over all neighbors of x and v
belongs to neighbors of x

DVR: Bellman-Ford Example (2) [6]



Clearly, $d_v(z) = 5$, $d_x(z) = 3$, $d_w(z) = 3$

B-F equation says:

$$\begin{aligned} d_u(z) &= \min \{ c(u,v) + d_v(z), \\ &\quad c(u,x) + d_x(z), \\ &\quad c(u,w) + d_w(z) \} \\ &= \min \{ 2 + 5, \\ &\quad 1 + 3, \\ &\quad 5 + 3 \} = 4 \end{aligned}$$

main diff is my immediate knowledge goes to router to help build global knowledge in router vs just sending to the neighbour(dvr)

Node that achieves minimum is next
hop in shortest path → forwarding table

Distance Vector Algorithm (3)

- $D_x(y)$ = estimate of least cost from x to y
- Distance vector: $\mathbf{D}_x = [D_x(y): y \in N]$
- Node x knows cost to each neighbor v : $c(x,v)$
- Node x maintains $\mathbf{D}_x = [D_x(y): y \in N]$
- Node x also maintains its neighbors' distance vectors
 - For each neighbor v , x maintains $\mathbf{D}_v = [D_v(y): y \in N]$

Distance Vector Algorithm (4)

Basic idea:

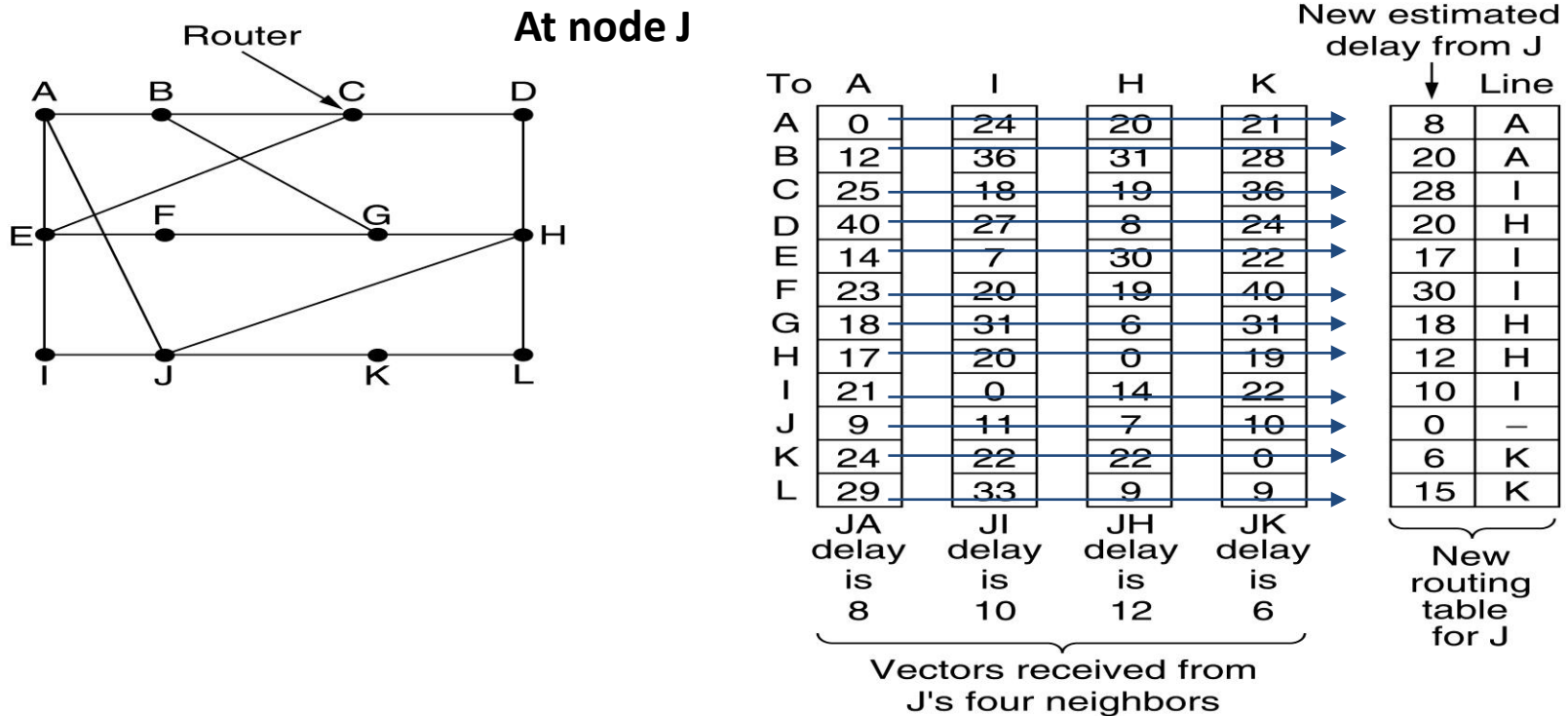
- Each node periodically sends its own distance vector estimate to neighbors v
- When a node x receives new DV estimate from neighbor, it updates its own DV using Bellman-Ford equation:

$$D_x(y) \leftarrow \min_v \{c(x, v) + D_v(y)\} \quad \text{for each node } y \in N, v \in \text{neighbor}$$

- Under minor, natural conditions, the estimate $D_x(y)$ converge the actual least cost $d_x(y)$

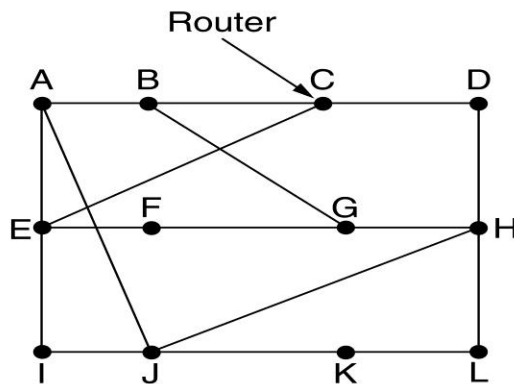
Distance Vector Routing (1) [2]

- Also known as **Bellman-Ford** routing
 - Used in **ARPANET**, till 1979
- Each router maintains a routing table**, with estimated “distance” to each destination (and **updates it periodically**)
- Each router periodically **exchanges this table with its neighbors**



Distance Vector Routing (2) [2]

- Each router measures “distance” on each outgoing link
 - Using e.g. queue length, round-trip delay
- It re-computes the routes as follows:



(a)

At node J

To	A	I	H	K	New estimated delay from J	
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	6	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	–
K	24	22	22	0	6	K
L	29	33	9	9	15	K

JA delay is 8	JL delay is 10	JH delay is 12	JK delay is 6
---------------	----------------	----------------	---------------

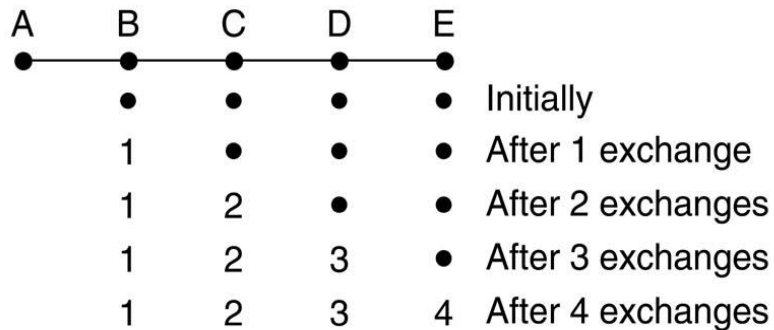
Vectors received from J's four neighbors

New routing table for J	
Line	
8	A
20	A
28	I
20	H
17	I
30	I
18	H
12	H
10	I
0	–
6	K
15	K

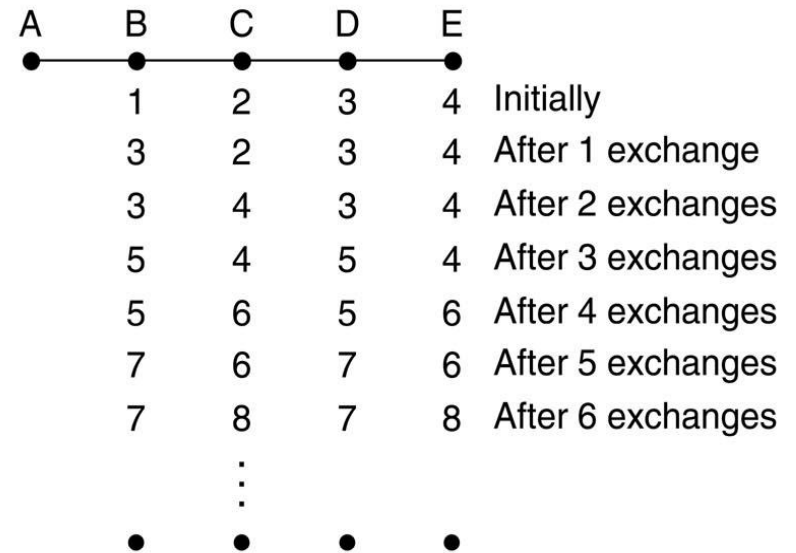
(b)

DVR: Count-to-infinity Problem [2]

- Several problems with Distance Vector routing:
 - Poor estimate of delays along each link
 - **Count-to-infinity problem:**
 - Good news spreads fast
 - Bad news travels slow, very slow

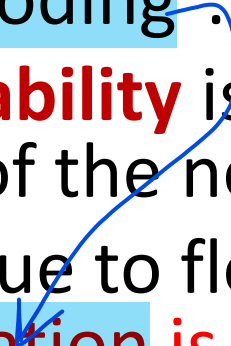


(a)



(b)

Link State Routing

- To mitigate the DVR Algo problems LSR was proposed.
 - It uses **Dijkstra Algorithm**.
 - Routing Tables contains the **Sequence No.** and **TTL** with other information.
 - Routing Information is exchanged with all the nodes using **"Flooding"**.
 - **Advantage: Reliability** is high as each node has the **global view** of the network
 - **Disadvantage:** Due to flooding and Packet Size, **Bandwidth utilization is very high.**
- 

DVR vs Link State Routing

- Distance Vector Routing:

- Tell neighbors about distance of all the destination
- Node's computation depends on neighbors
- Each router maintains distance vector, (dist , cost) tuple per destination
- Periodically send copy of distance vector to all neighbors
- DV node can advertise incorrect *path* cost
- each node's table used by others that leads to error propagate through network

- Link State Routing :

- node can advertise incorrect *link* cost
- each node computes only its *own* table
- Tell about distance to each neighbor to all routers
- Each router computes its best paths

Complexity of LS vs DV Algorithms [6]

Message complexity

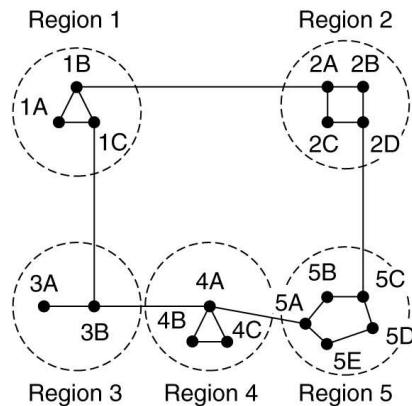
- LS: with n nodes, E links, $O(nE)$ msgs sent
- DV: exchange between neighbors only

Speed of Convergence

- LS: $O(n^2)$ algorithm requires $O(nE)$ msgs
- DV: convergence time varies
 - may be routing loops
 - count-to-infinity problem

Hierarchical Routing [2]

- Is it possible to govern the whole N/w using one **Routing Algo**?
- Essentially solves “scalability” problem for large networks
- Considers a network to consist of a connected network of regional networks
- Routing is either within the local region, or across regions
- Multiple levels of hierarchy (2 or more)



(a)

Full table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

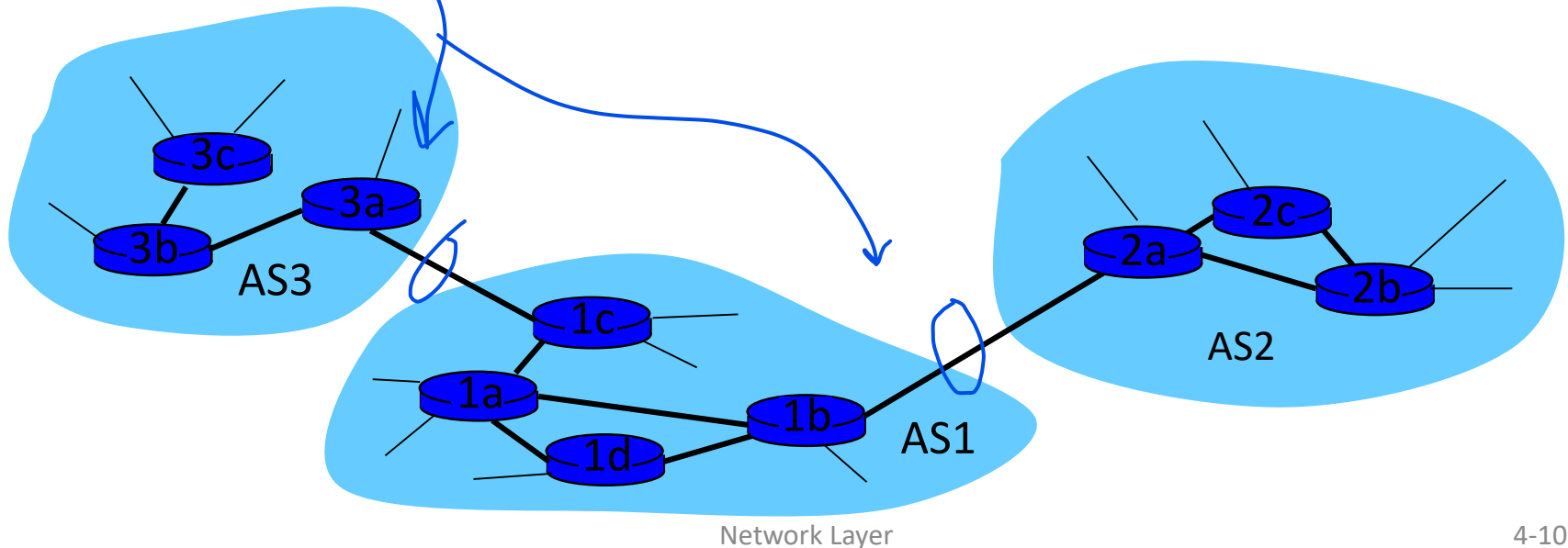
Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

(c)

Internet Routing [6]

Example: Choosing among multiple ASes

- Now suppose AS1 learns from the inter-AS protocol that subnet **x** is reachable from AS3 *and* from AS2.
- To configure forwarding table, router 1d must determine towards which gateway it should forward packets for destination **x**.
- This is also the job on inter-AS routing protocol!

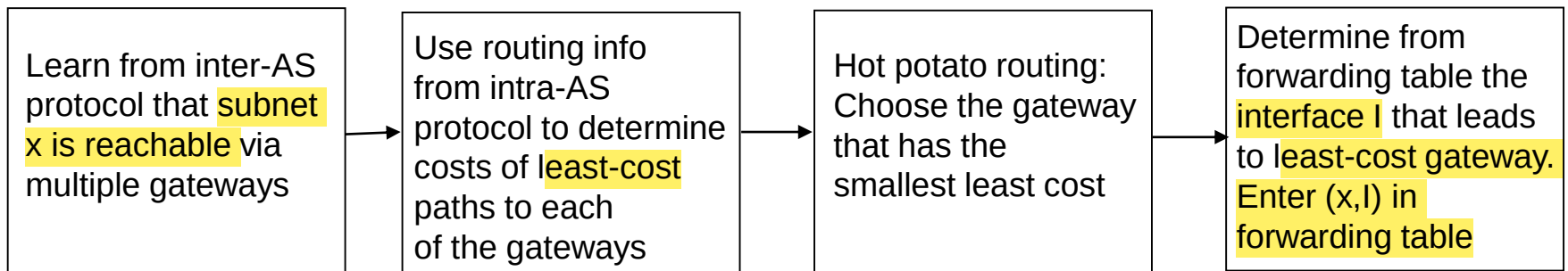


Internet Routing [6]

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- To configure forwarding table, router 1d must determine towards which gateway it should forward packets for destination **x**.
- This is also the job on inter-AS routing protocol!
- **Hot potato routing:** send packet towards closest of two routers.

reachable via inter-AS , routing info least , hot potato , least cost gateway via forward table

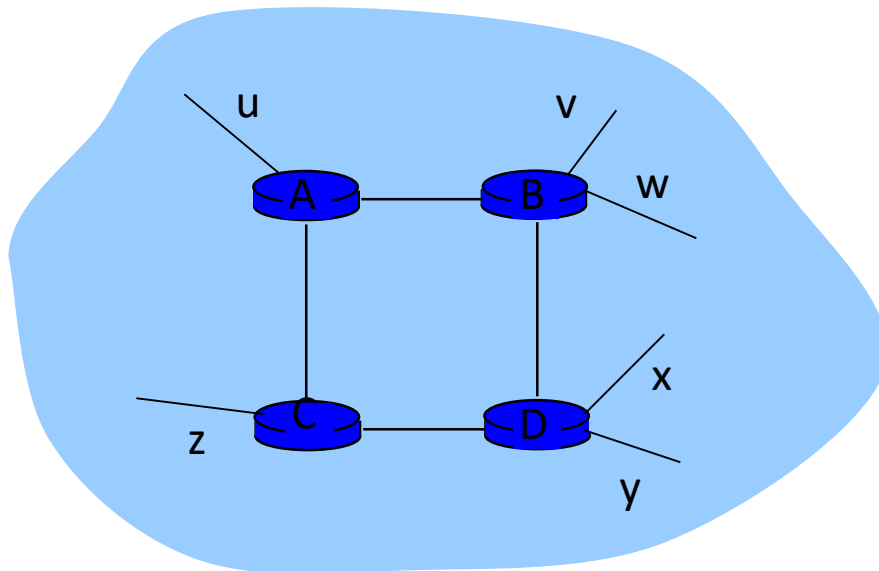


Intra-AS Routing

- Also known as **Interior Gateway Protocols (IGP)**
- Most common Intra-AS routing protocols:
 - RIP: Routing Information Protocol
 - OSPF: Open Shortest Path First
 - IGRP: Interior Gateway Routing Protocol (Cisco proprietary)

RIP (Routing Information Protocol) [6]

- Distance vector algorithm is used by RIP.
- Included in BSD-UNIX Distribution in 1982
- Distance metric: # of hops (max = 15 hops)
- Each hop cost is 1 (Fixed)



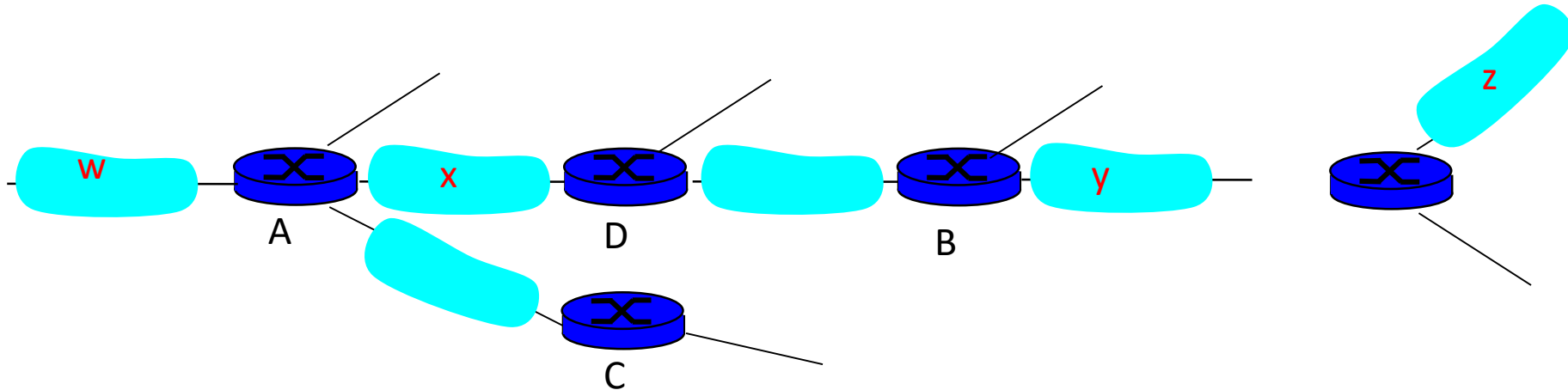
A's Table

<u>destination</u>	<u>hops</u>
u	1
v	2
w	2
x	3
y	3
z	2

RIP advertisements

- **Distance vectors:** exchanged among neighbors every 30 sec via Response Message (also called **advertisement**)
- **Each advertisement:** list of up to 25 destination within AS.

RIP: Example [6]



Destination Network	Next Router	Num. of hops to dest.
w	A	2
y	B	2
z	B	7
x	--	1
....		

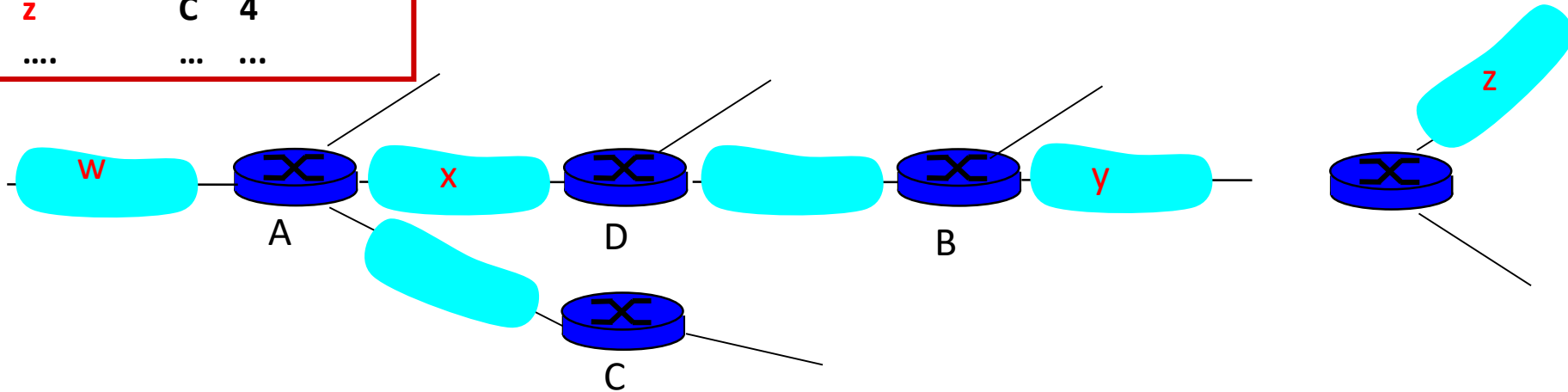
Routing table in D

RIP: Example [6]

see

Advertisement
from A to D

Dest	Next hops	
w	-	-
x	-	-
z	C	4
...	...	...



Destination Network	Next Router	Num. of hops to dest.
w	A	2
y	B	2
z	B A	7 5
x	-	1
....		

Routing table in D

RIP: Link Failure and Recovery

If no advertisement heard after **180 sec** --> neighbor/link declared dead

- routes via neighbor invalidated
- new advertisements sent to neighbors
- neighbors in turn send out new advertisements (if tables changed)
- link failure info quickly propagates to entire network.
- Message in reverse direction (poison reverse) is used to prevent ping-pong loops (infinite distance = 16 hops)

OSPF (Open Shortest Path First)

- **Successor** of RIP Protocol.
- “Open”: publicly available
- Uses **Link State algorithm**
 - LS packet dissemination
 - Topology map at each node
 - Route computation using **Dijkstra’s algorithm**
- OSPF advertisement carries one entry per neighbor router
- Advertisements disseminated to **entire AS** (via flooding)
 - Carried in **OSPF messages directly over IP** (rather than **TCP or UDP**)

OSPF (Open Shortest Path First)

- **Network Administrator** will decide the Link Cost
 - Most of the time the reverse of the “Link Capacity”.
- Every **30 Minutes** the routing information is exchanged.

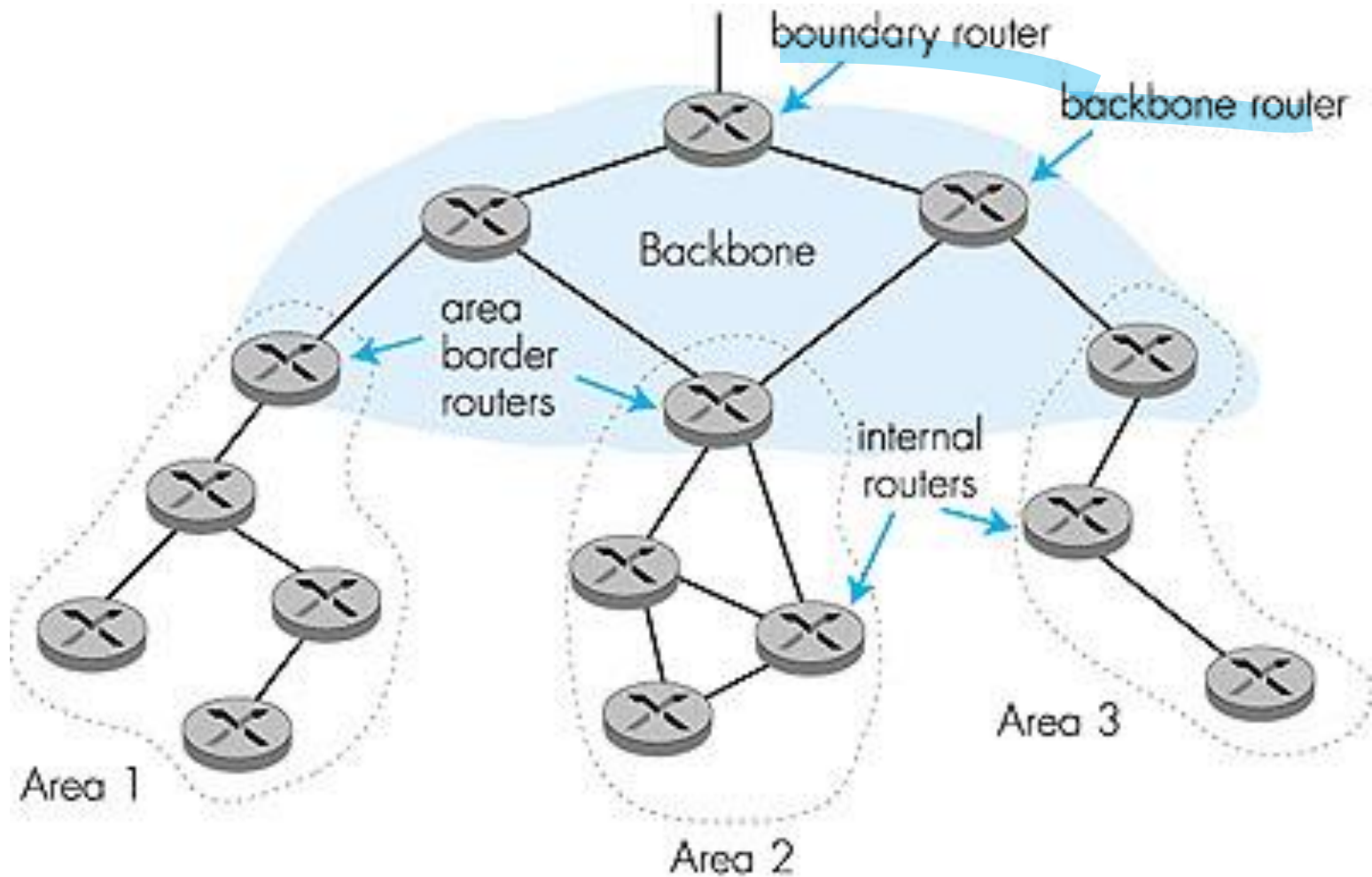
OSPF “advanced” features (not in RIP)

- **Security:** all OSPF messages **authenticated** (to prevent malicious intrusion)
 - **Simple** – Text
 - **MD5** – Hash Key
- **Multiple** same-cost **paths** allowed (only one path in RIP).
 - All the same cost path would be utilized, So the **congestion** can be avoided. 
- Integrated **unicast** and **multicast** support:
 - **Unicast** we have all the **information**.
 - **Multicast OSPF (MOSPF) uses same topology data base as OSPF**

OSPF “advanced” features (not in RIP)

- **Hierarchical** OSPF in large domains.
 - Inside the autonomous system Hierarchy is supported.
 - AS is divided in to **Areas**
 - **Area Boarder Router**
 - **Backbone Area**

Hierarchical OSPF [6]



Hierarchical OSPF

- **Two-level hierarchy:** local area, backbone.
 - Link-state advertisements only in area
 - each nodes has detailed area topology; only know direction (shortest path) to nets in other areas.
- **Area border routers:** “summarize” distances to nets in own area, advertise to other Area Border routers.
- **Backbone routers:** run OSPF routing limited to backbone.
- **Boundary routers:** connect to other AS's.

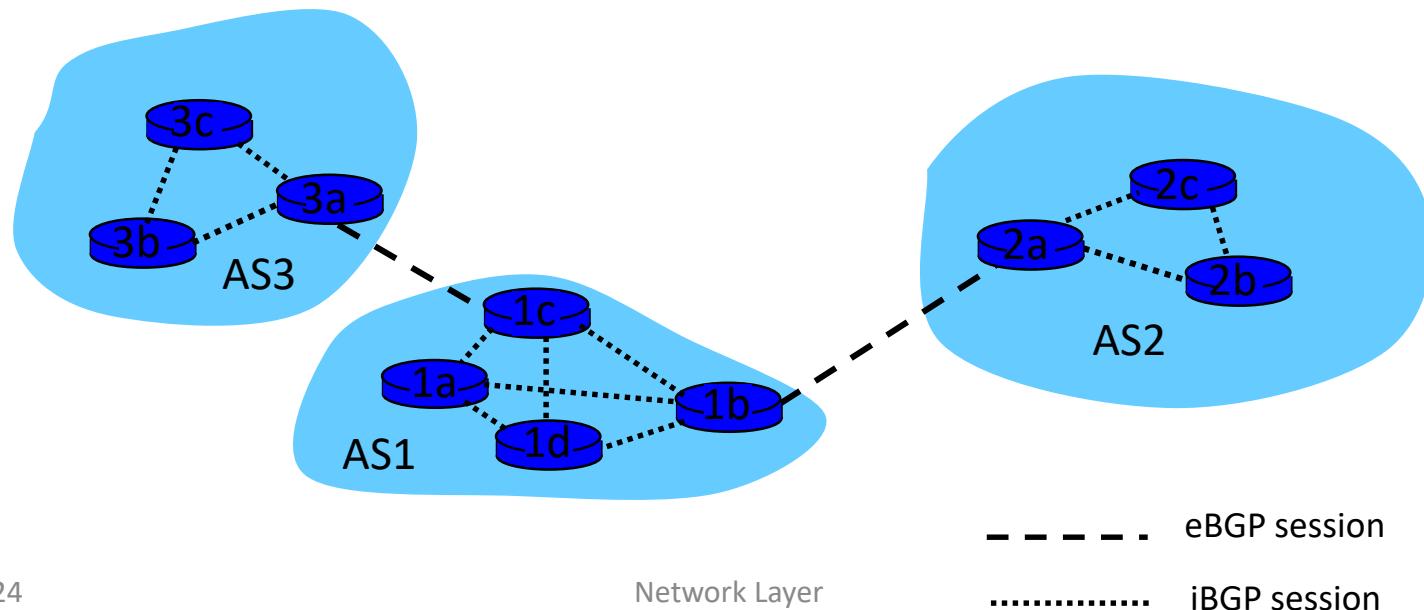
Internet inter-AS routing: BGP

- **BGP (Border Gateway Protocol):** *the de facto standard*
- BGP provides each AS a means to:
 1. Obtain **subnet reachability information** from neighboring ASs.
 2. Propagate the reachability information to **all routers internal to the AS.**
 3. Determine **“good” routes** to subnets based on reachability information and policy.
- Allows **a subnet** to advertise its existence to rest of the Internet: ***“I am here”***

BGP Basics [6]

- Pairs of routers (**BGP peers**) exchange routing info over **semi-permanent TCP connections**: BGP sessions (eBGP, iBGP)
- Note that BGP sessions do not correspond to physical links.
- When AS2 advertises a prefix to AS1, AS2 is *promising* it will forward any datagram's destined to that prefix towards the prefix.
 - AS2 can **aggregate prefixes** in its advertisement

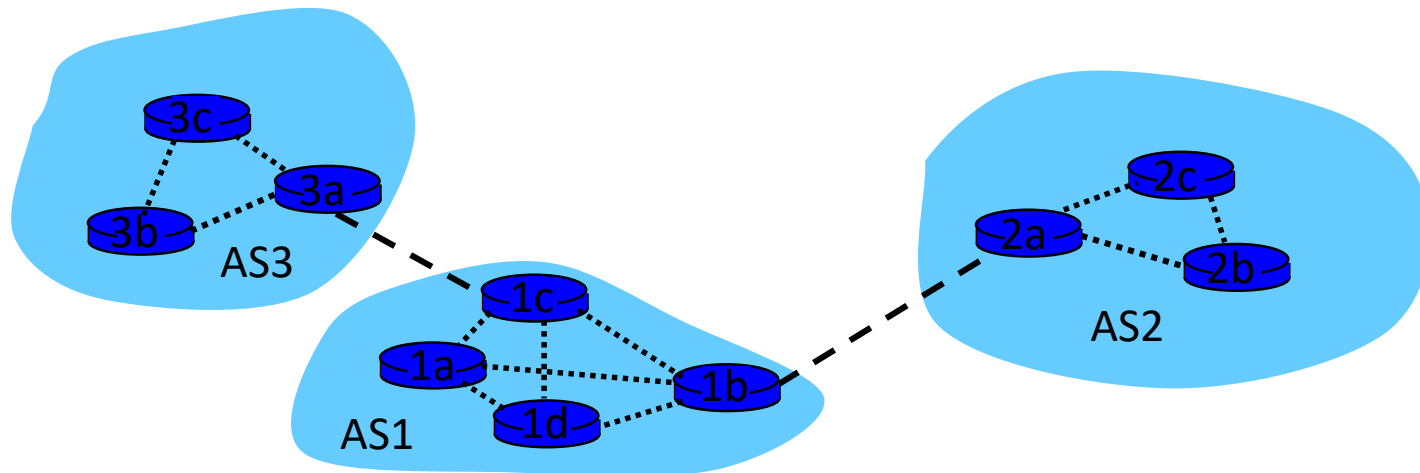
bgp peers , semio tcp , as advertise towards prefix



Distributing Reachability Info[6]

- With **eBGP (external)** session between 3a and 1c, AS3 sends prefix reachability info to AS1.
- 1c can then use **iBGP (internal)** to distribute this new prefix reach info to all routers in AS1
- 1b can then **re-advertise** the new reach info to AS2 over the **1b-to-2a eBGP session**
- **Subnets in BGP** are called **Prefix**.
- When router learns about a new prefix, it creates an entry for the prefix in its forwarding table.

eBGP to advertise inter AS reachability , then particular 1b shares via iBGP to in As nodes and eBGP for outer AS



Path attributes & BGP routes

- When advertising a prefix, advertisement includes BGP attributes.
 - prefix + attributes = “route”
- **Two important attributes:**
 - **AS-PATH:** contains the ASs through which the advertisement for the prefix passed: AS 67 AS 17
 - **NEXT-HOP:** Indicates the specific internal-AS router to next-hop AS. (There may be multiple links from current AS to next-hop-AS.) – Recall **Hot Potato Algo**
- When gateway router receives route advertisement, uses **import policy** to accept/decline. (**More than one path is possible**)

BGP Route Selection [6]

- Router may learn about **more than 1 route to some prefix**. Router must select a route.
- **Elimination rules:**
 1. **Local preference value** attribute: **policy decision**
 2. **Shortest AS-PATH**
 3. **Closest NEXT-HOP** router: **hot potato routing**
 4. Additional criteria

(20 --> 10 --> 4 --> 1)

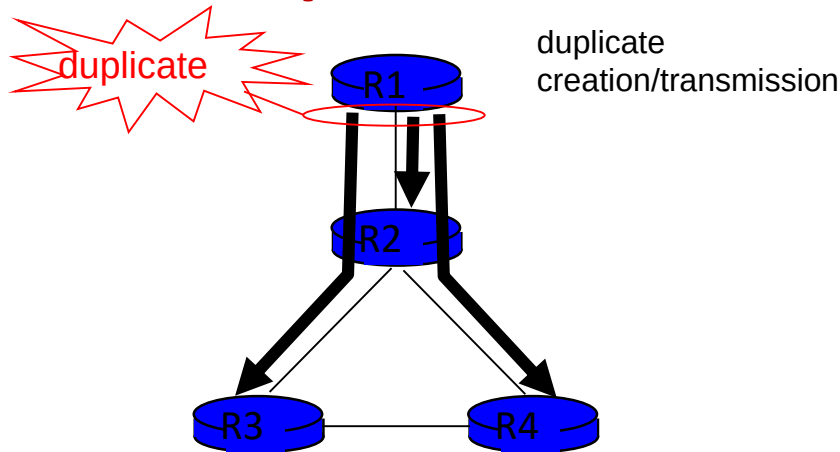
BGP Messages [6]

- BGP messages exchanged using TCP.
- BGP messages:
 - **OPEN:** opens TCP connection to peer and authenticates sender
 - **UPDATE:** advertises new path (or withdraws old)
 - **KEEP ALIVE:** keeps connection alive in absence of UPDATES; also ACKs OPEN request
 - **NOTIFICATION:** reports errors in previous msg; also used to close connection

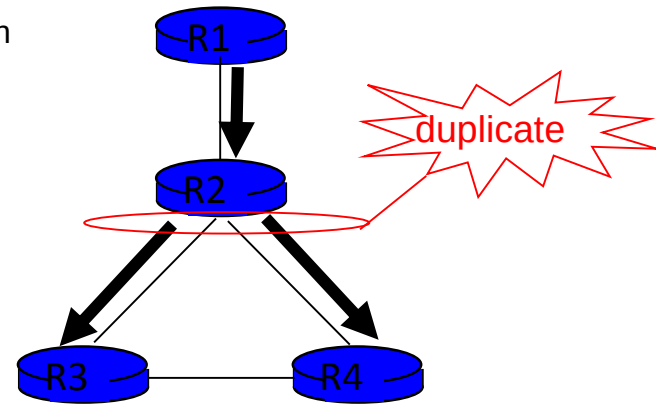
oukn

Broadcast Routing

- Deliver packets from **source to all other nodes**
- **Source duplication** is inefficient:



Source duplication (**N-way Unicast**)



In-network duplication (**Flooding**)

❑ **Source duplication:** Source need to know the addresses of all the recipients

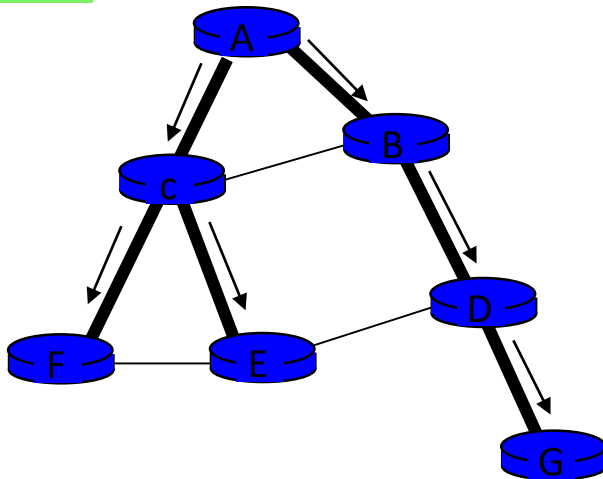
❑ Not Realistic, **Congestion** Problem

In-network duplication

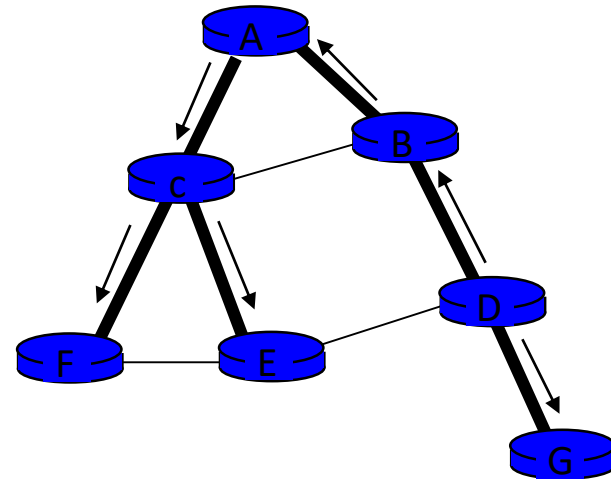
- **Flooding:** when node receives broadcast packet, sends copy to all neighbors
 - **Problems:** cycles & broadcast storm
- **Controlled flooding:** node only broadcast packet if it hasn't broadcast same packet before
 - Node keeps track of **packet ids** already broadcasted
 - Or **reverse path forwarding (RPF):** only forward packet if it arrived on shortest path between node and source
- **Spanning tree**
 - No redundant packets received by any node

Spanning Tree

- First **construct a spanning tree**
- Nodes forward copies only along spanning tree



(a) Broadcast initiated at A



(b) Broadcast initiated at D

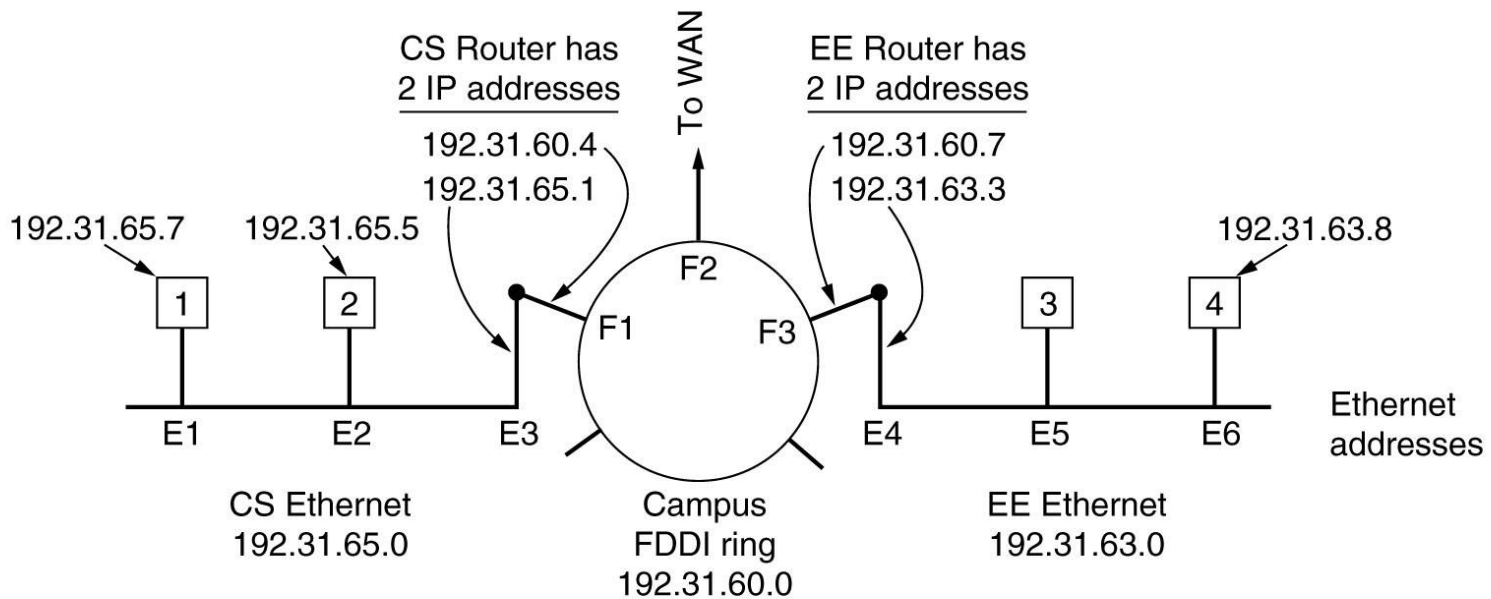
Internet Control Protocols [2]

- Several protocols:
 - ARP, RARP (these are discussed later)
 - ICMP (Internet Control Message Protocol)
 - Several messages, including “echo” and “echo-reply” used to “ping” hosts
 - ICMP packets will be encapsulated inside an IP packet

Message type	Description
Destination unreachable	Packet could not be delivered
Time exceeded	Time to live field hit 0
Parameter problem	Invalid header field
Source quench	Choke packet
Redirect	Teach a router about geography
Echo	Ask a machine if it is alive
Echo reply	Yes, I am alive
Timestamp request	Same as Echo request, but with timestamp
Timestamp reply	Same as Echo reply, but with timestamp

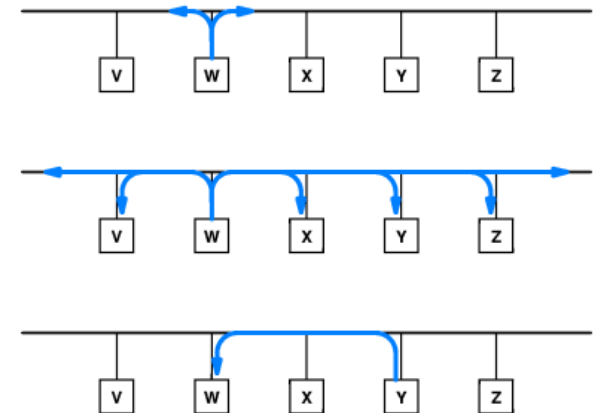
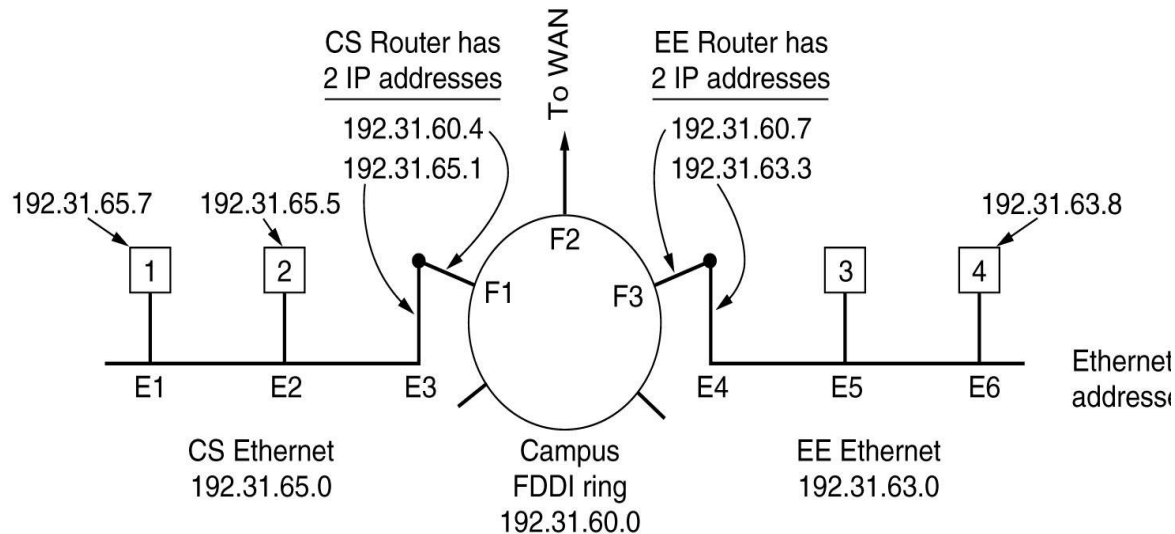
ARP Protocol [2]

- ARP protocol: “address resolution protocol”
 - IP address $\leftrightarrow$ Data-link (or physical) address
 - This is distinct from “domain-name” $\leftrightarrow$ IP address problem



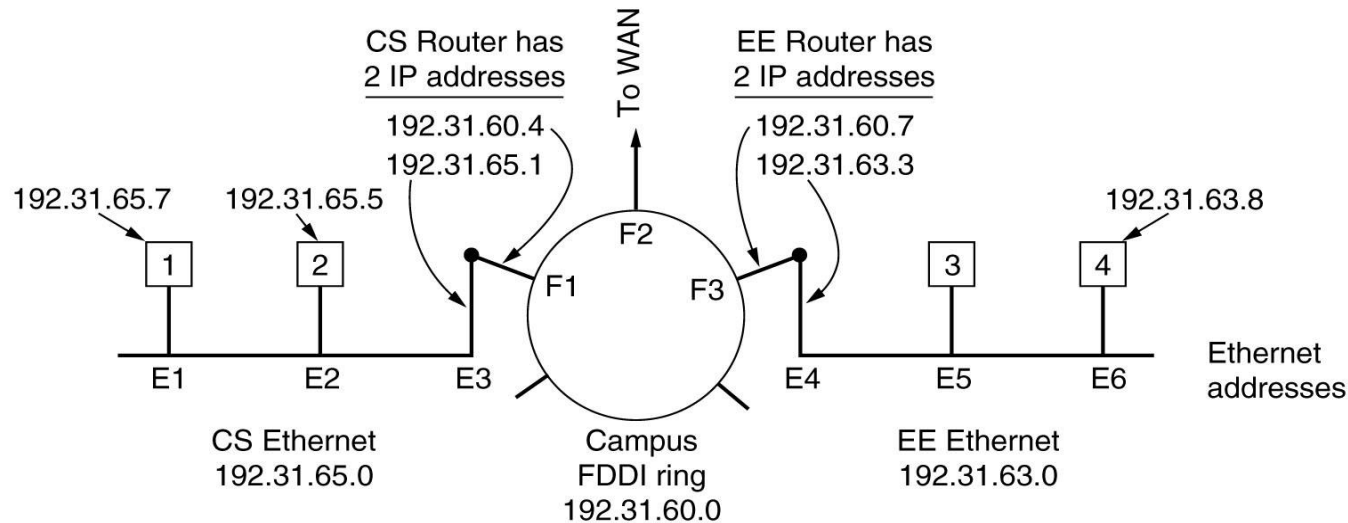
ARP Protocol(2) [2]

- ARP protocol:
 - ARP-REQ ARP-REPLY packets
 - ARP-REQ is broadcast over local subnet only
 - Destination IP address $\leftrightarrow$ Ethernet address is cached by source, once a reply is received
 - The destination also caches similar info about the source



ARP Protocol(3) [2]

- Consider H1 to H4 communication
 - H1 issues an ARP-REQ, to which CS router responds with its E3 address
 - CS router issues an ARP-REQ on FDDI ring, to which EE router responds with its F3 address
 - EE router issues an ARP-REQ on the Ethernet, to which H4 responds with its E6 address



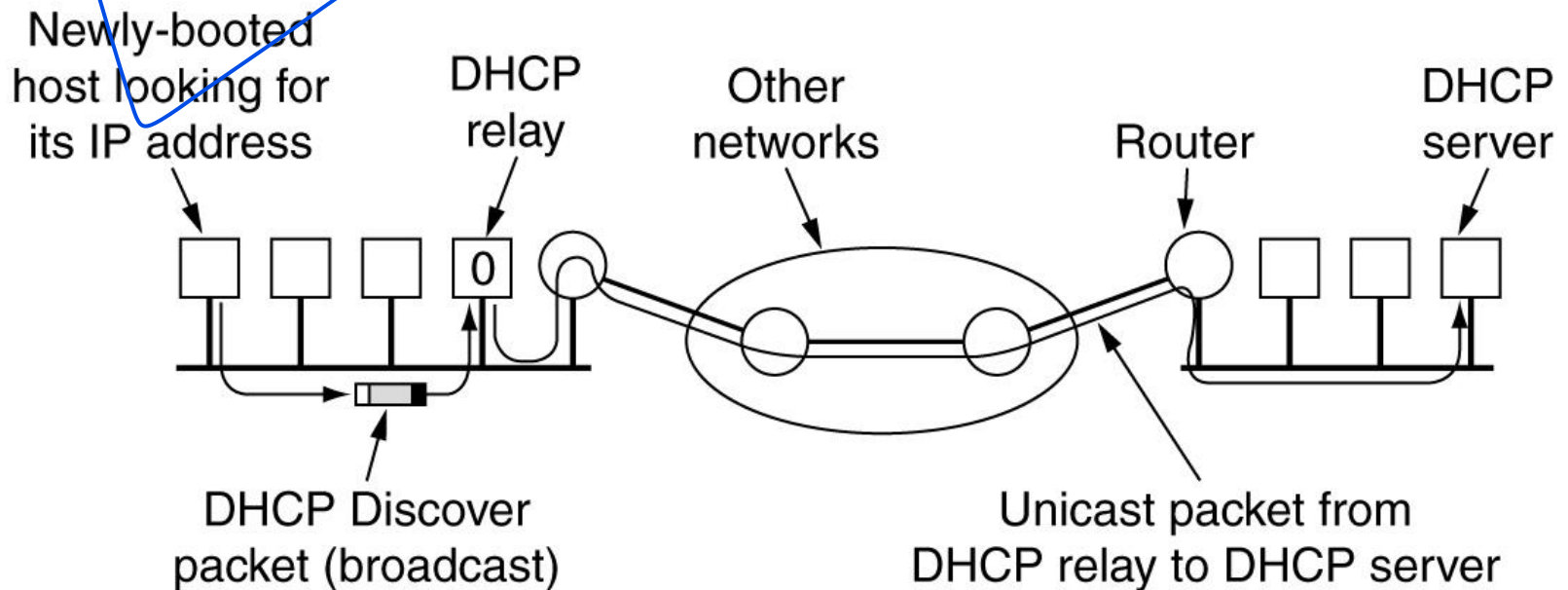
RARP Protocol

router , ip table , where , dhcp

- ARP gives IP-addr → Physical-addr
- RARP solves the problem of “what is my IP address”?
 - A problem that occurs in disk-less workstations, that have no disk resident OS
- RARP-REQ issued by client, while RARP-REPLY is sent by RARP server
- Need a RARP server for each network separated by a router
- BOOTP - It also provides a diskless workstation with additional information
 - including the IP address of the file server holding the memory image and the IP address of the default router
- Both problems solved using DHCP protocol

Dynamic Host Configuration Protocol [2]

Operation of DHCP.



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Thank You